
Beyond Incremental Beam Search for Compositional Explanations of Neurons

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Abstract

Compositional explanations describe the alignment between neuron activations and concepts through logical formulas, obtained by solving a search problem over combinations of concepts. Due to the combinatorial nature of this space, prior work relies on incremental beam search, which restricts the exploration but limits the expressiveness of explanations and may discard intermediate candidates needed to identify higher alignment. In this paper, we propose a non-incremental beam search framework that expands the search space while remaining empirically tractable. Our approach introduces a non-incremental beam expansion strategy and a functional-equivalence-aware beam search that leverages equivalence classes of formulas to reduce redundant beam entries and improve search quality. Experiments on natural language and vision tasks show that our method finds explanations expressing a higher alignment than the incremental formulation while maintaining comparable runtime, and leads to a significant increase in the number of optimal explanations when these can be computed. Additionally, these explanations exhibit higher perceived descriptive power while preserving understandability. Overall, these results show that expanding search beyond incremental formulations leads to more expressive and accurate explanations.

1 Introduction

Compositional explanations [26] are a class of methods that describe the alignment between neuron activations and human-interpretable concepts through propositional logic formulas. These explanations capture how neurons respond to combinations of low-level (e.g., colors, textures) and high-level (e.g., objects, semantic attributes) concepts by identifying logical rules that best match the spatial or feature-level activation patterns of a neuron. Compositional explanations are typically obtained by solving a search problem over the space of possible concept combinations, where the objective is to identify the formula that maximizes alignment with the neuron activation mask.

A major problem in this setting is that the search space grows combinatorially with the number of concepts and the length of the explanation. As a result, prior work has adopted strong assumptions on the formula structure and leverages beam search to restrict the exploration to a subset of the state space [26, 12, 23, 22, 21]. In particular, the current literature enforces an incremental construction of formulas, where concepts are added sequentially. Although this design makes the search efficient, it also limits the expressiveness of the resulting explanations and may prevent the discovery of formulas that express higher alignment between activations and concepts (Figure 1). Indeed, intermediate candidates that are not selected by beam search at early stages cannot be revisited later, even if they would lead to better explanations when combined differently.

In this paper, **we take a first step towards addressing the limitations imposed by incremental beam search**. Specifically, we propose a **non-incremental** beam search framework that expands the exploration beyond the incremental formulation while remaining empirically tractable. Our

Incremental Beam Search $(((((hyp:standing \wedge pre:sitting) \wedge hyp:tag:.) \vee hyp:though) \wedge \neg hyp:tag:rb)$ IoU: 0.218	Our Approach: Non-Incremental Beam Search $((hyp:standing \vee neighbor:hyp:runner) \wedge ((pre:sitting \vee pre:sits) \vee pre:sit))$ IoU: 0.368												
Top 5-activating samples													
P: A little girl sitting on a toy tractor. H: The girl is standing . P: A girl lays on a wooden deck. H: A person is running . P: A little boy is sitting next to a bed. H: A boy is jumping on the bed. P: Women and children sitting in the shade. H: The woman is running . P: A gray-haired woman sitting on a bench. H: The woman is running .	<table> <thead> <tr> <th>Incremental</th> <th>Ours</th> </tr> </thead> <tbody> <tr><td>●</td><td>●</td></tr> <tr><td>●</td><td>●</td></tr> <tr><td>●</td><td>●</td></tr> <tr><td>●</td><td>●</td></tr> <tr><td>●</td><td>●</td></tr> </tbody> </table>	Incremental	Ours	●	●	●	●	●	●	●	●	●	●
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Figure 1: An example of explanations computed by incremental beam search and our non-incremental approach for a neuron that recognizes contradictions between sitting actions and standing or moving actions.

approach introduces a novel beam expansion strategy that generates candidates by combining the best candidate formulas previously explored and by inserting concepts within internal substructures, thus enabling the search to access regions of the state space that are unreachable under the incremental assumption. Additionally, we propose a functional-equivalence-aware beam search that identifies labels that induce the same activation mask and leverages this property to prevent redundant formulas from occupying multiple beam positions, thus increasing the diversity of beam candidates. We evaluate the proposed framework on both natural language inference and image classification tasks across 6 different models, following standard setups for this area of research.

In summary, our contributions are as follows:

- We introduce the notion of functional equivalence for compositional explanations, formalizing when different logical formulas induce the same activation mask.
- We propose a non-incremental beam search framework that supports the generation of non-incremental formulas and explicitly manages functional equivalence during beam expansion.
- We show that the proposed framework improves explanation alignment across natural language and vision tasks, more frequently recovers optimal explanations, and does not introduce a loss in understandability while improving perceived descriptive power.

The code associated with this paper is available at https://github.com/aiea-lab/nonincremental_ce

2 Related Work

Neuron explanations aim to understand what individual neurons learn during training. A wide range of methods has been proposed across both vision and natural language domains [24], such as approaches that generate representative samples that maximize neuron activations [9, 30, 27], methods that associate neurons with concepts or textual descriptions through correlation-based analyses and foundational models [14, 28, 29, 16, 1, 33, 11], and approaches that study internal representations through probing or sparse decompositions [8, 10, 6].

Among these, compositional explanations [26] represent a specific family of methods that focus on the **alignment between a neuron activation range and the locations of concepts**, and that express this relationship through propositional logic formulas. The seminal work in this area is Network Dissection [3, 4], which associates each neuron with the single concept that maximizes alignment. This approach was extended by compositional explanations [26], which consider combinations of concepts to model polysemantic behavior, where neurons respond to multiple, potentially unrelated concepts. Subsequent work has extended this formulation to capture richer relationships between concepts, including co-occurrence [3], exclusion [26], relative position [12], and hierarchy [23], to capture behavior across multiple activation ranges [21], to broaden its applicability [19], and to provide formal guarantees about the optimality of the explanations [20].

This paper targets the incremental construction assumption of compositional explanation methods. While this assumption is effective in practice, it limits the set of formulas that can be explored and

may prevent the discovery of the best alignment. Our work addresses this limitation by proposing a non-incremental search framework that expands the beam state space while remaining empirically tractable, and by explicitly modeling functional equivalence between formulas during the search. Our proposal shares some conceptual ideas with literature on combinatorial search over symbolic expressions. Similarly to our algorithm, in Inductive Logic Programming, logical hypotheses are explored through refinement operators [32] that progressively transform candidate clauses, while program synthesis methods can prune candidate programs that behave identically on the available examples through observational equivalence [2]. Related ideas also appear in genetic programming, where larger expressions are constructed by recombining previously generated subexpressions [17]. These approaches share with our work the need to search a large structured space while avoiding redundant candidates, but address different settings. Our contribution applies these ideas to the interpretability settings through the concepts of recombination and functional-equivalence.

3 Method

This section introduces the compositional explanation problem, the notation, and terminology used in the area. Then, we present the concept of functional equivalence and describe the proposed non-incremental beam search algorithm.

Background and Terminology Let $\mathcal{D} = \{x_1, \dots, x_m\}$ be a probing dataset used to analyze the behavior of a trained model. For each sample $x \in \mathcal{D}$, let Ω_x denote the set of locations and $\Omega = \bigcup_{x \in \mathcal{D}} \Omega_x$. A location may correspond to the whole sample or to a specific element within the sample, depending on the considered setting. Let $\mathcal{L}^1 = \{c_1, \dots, c_p\}$ be a set of atomic concepts. Each concept $c_k \in \mathcal{L}^1$ is associated with a binary mask $M_k \subseteq \Omega$ representing the locations where the concept is present. We denote by $\mathbf{M} = \{M_k\}_{k=1}^p$ the collection of all concept masks. Let $\mathcal{O}$ be the set of all admissible logical connectives. In this work, in line with prior work, we consider as connectives $\wedge$, $\vee$, and $\wedge \neg$, where $\wedge \neg$ denotes conjunction with negation applied to an atomic concept. Given a neuron z , we define its activation mask as $N \subseteq \Omega$ representing the locations where the neuron activates within a selected activation range $[\tau_1, \tau_2]$. The activation range can be identified by thresholding the highest activations [26] or by using clustering techniques [21]. When the dimensions of the activation space and input differ, we follow [26] and apply interpolation to project the activations into Ω . Let $\mathcal{L}^n$ denote the set of admissible logical formulas of arity at most n constructed from $\mathcal{L}^1$, and let $C(L)$ denote the set of atomic concepts appearing in the label L . Each label $L \in \mathcal{L}^n$ is associated with a semantic operator

$$\circ_L(\mathbf{M}) \subseteq \Omega \quad (1)$$

defined recursively by applying logical connectives to concept masks. In particular, for $L = c_k$, $\circ_L(\mathbf{M}) = M_k$, and for composite labels:

$$\circ_{L_1 \wedge L_2}(\mathbf{M}) = \circ_{L_1}(\mathbf{M}) \cap \circ_{L_2}(\mathbf{M}) \quad \circ_{L_1 \vee L_2}(\mathbf{M}) = \circ_{L_1}(\mathbf{M}) \cup \circ_{L_2}(\mathbf{M}) \quad \circ_{L_k \wedge \neg c_j}(\mathbf{M}) = \circ_{L_k}(\mathbf{M}) \setminus M_j$$

Finally, let $L = A \wedge \neg c$, where $c \in \mathcal{L}^1$. The occurrence of $\neg c$ is *unverified* if $\circ_{A \wedge \neg c}(\mathbf{M}) = \circ_A(\mathbf{M})$. A label is *unverified* if it contains at least one unverified negation.

The goal of compositional explanations is to identify

$$L^* = \arg \max_{L \in \mathcal{L}^n} IoU(N, \circ_L(\mathbf{M})) \quad (2)$$

where the IoU score is defined as

$$IoU(A, B) = \frac{|A \cap B|}{|A \cup B|} \quad (3)$$

and measures the overlap between the neuron activation mask and the mask induced by a candidate explanation, normalized by their union. The higher values indicate stronger alignment between the neuron activations and the explanation.

Given the IoU objective and the fixed admissible space $\mathcal{L}^n$, any L^* satisfying Equation (2) is an *optimal solution* [20]. Because computing such a solution is often impractical or prohibitively expensive [26], previous methods restrict the search space by two assumptions: each concept may appear at most once in a label, and labels are constructed incrementally by adding one concept at a time

to the current formula. Under these assumptions, labels take the form $((c_1 \oplus_1 c_2) \oplus_2 c_3) \dots \oplus_{k-1} c_k$, where $c_i \in \mathcal{L}^1$ are distinct concepts and $\oplus_i \in \mathcal{O}$. Even with these restrictions, the search space grows rapidly with the vocabulary size. For this reason, prior work relies on beam search to make the problem tractable. However, the combination of incremental construction and beam search can limit the complexity of the relationships captured by the explanations, thus limiting its expressiveness. Our method addresses this limitation by expanding the beam search with additional non-incremental candidates and by explicitly accounting for functional equivalence among labels.

Running example As a running example throughout this and the following sections, consider a probing dataset containing five sentences: $\mathfrak{D} = \{x_1 : \text{A funny child is sleeping on the couch.}, x_2 : \text{A funny man is standing in a quiet room.}, x_3 : \text{A dog is sleeping outdoors.}, x_4 : \text{A child is sleeping outdoors in a quiet park.}, x_5 : \text{A woman is sleeping on the sofa.}\}$. Consider the atomic concepts $F = \text{funny}$, $Q = \text{quiet}$, $S = \text{sleeping}$, and $O = \text{outdoors}$. Their concept masks are:

$$M_F = [1, 1, 0, 0, 0] \quad M_Q = [0, 1, 0, 1, 0] \quad M_S = [1, 0, 1, 1, 1] \quad M_O = [0, 0, 1, 1, 0]$$

since, for example, funny appears in x_1 and x_2 but not in the remaining samples. Similarly, if the neuron activates within the considered range for x_1 and x_3 , we can write the activation mask as:

$$N = [1, 0, 1, 0, 0]$$

In this example, a formula such as $L_1 = (F \vee Q) \wedge S$, induces a mask $\circ_{L_1}(M) = [1, 0, 0, 1, 0]$, and therefore an alignment of $IoU(N, \circ_{L_1}(M)) = 1/3 \approx 0.33$.

3.1 Functional Equivalence

We propose and define functional equivalence as an equivalence relation over labels based on their induced masks. Indeed, we observe that, in general, multiple syntactically different labels may induce the same mask over Ω . We formalize this notion through functional equivalence.

Definition 3.1 (Functional Equivalence). Two labels $L_1, L_2 \in \mathcal{L}^n$ are functionally equivalent if $\circ_{L_1}(\mathbf{M}) = \circ_{L_2}(\mathbf{M})$ or, equivalently, $IoU(\circ_{L_1}(\mathbf{M}), \circ_{L_2}(\mathbf{M})) = 1$.

Functional equivalence partitions $\mathcal{L}^n$ into equivalence classes. We distinguish three mutually exhaustive cases: logical equivalence (Type 1), dataset equivalence (Type 2), and dataset ambiguity (Type 3). The first two arise between formulas sharing the same concepts, while the third involves formulas differing in their concepts.

Type 1: Logical Equivalence

$$L_1 \equiv_{\text{logic}} L_2 \iff \forall \mathbf{M}, \circ_{L_1}(\mathbf{M}) = \circ_{L_2}(\mathbf{M})$$

This equivalence is intrinsic to the logical structure of the formulas (e.g., commutativity) and holds independently of the dataset. In practice, checking logical equivalence is computationally expensive and, in prior work, is not explicitly enforced during search beyond the simplest cases.

Type 2: Dataset Equivalence

$$L_1 \equiv_{\mathfrak{D}} L_2 \iff \circ_{L_1}(\mathbf{M}) = \circ_{L_2}(\mathbf{M}) \wedge \mathcal{C}(L_1) = \mathcal{C}(L_2) \wedge L_1 \not\equiv_{\text{logic}} L_2$$

This occurs when two labels use the same concepts but differ syntactically, yet produce identical masks due to dataset-specific properties (e.g., absence of certain co-occurrences). For example, in the running example, consider $L_1 = F \vee (Q \wedge S)$ and $L_2 = (F \wedge S) \vee Q$. The two labels contain the same concepts but are not logically equivalent. However, both induce the mask $[1, 1, 0, 1, 0]$ on the probing dataset, and are therefore Type-2 equivalent.

Type 3: Dataset Ambiguity.

$$L_1 \equiv_{\text{amb}} L_2 \iff \circ_{L_1}(\mathbf{M}) = \circ_{L_2}(\mathbf{M}) \wedge \mathcal{C}(L_1) \neq \mathcal{C}(L_2)$$

Unlike Type 2, this case involves different concept sets and therefore captures ambiguity induced by the probing dataset. This ambiguity can be global, when different concepts or formulas induce the same mask over the dataset, or context-dependent, when they become indistinguishable only within a shared subformula. For example, in the running example, O and $S \wedge O$ are functionally equivalent since $\circ_O(M) = M_O = [0, 0, 1, 1, 0] = M_S \cap M_O = \circ_{S \wedge O}(M)$.

3.2 Non-Incremental Algorithm

To move beyond incremental approaches, our algorithm proposes a novel design for beam expansion and a functional-equivalence-aware beam search.

3.2.1 Beam Expansion

Let $k + 1$ be the current beam level, let B_k be the set of labels selected for expansion (i.e., the set of top b labels in the beam at the previous step k), and let $B_{\leq k} = \bigcup_{r=1}^k B_r$ be the set of labels selected for expansion up to and including level k . For example, in the running example, $B_{\leq k}$ may contain $L_1 = (F \vee Q) \wedge S \in B_3$, $L_2 = F \vee Q \in B_2$, and $L_3 = S \wedge O \in B_2$. Given a label $L \in B_k$ and an atomic concept $c \in \mathcal{L}^1$ not already appearing in L , our expansion procedure generates three families of candidates.

Incremental candidates. The first family corresponds to the standard incremental expansion used in prior beam-search approaches. For each admissible operator $\oplus \in \mathcal{O}$, the algorithm generates

$$L_{\text{inc}} = L \oplus c \quad (4)$$

For example, incrementally adding O to L_1 through $\wedge$ produces $C_{\text{inc}} = ((F \vee Q) \wedge S) \wedge O$, whose induced mask in the running example is $[0, 0, 0, 1, 0]$.

Non-incremental candidates. The second family changes the scope of a newly added concept by inserting it inside one of the internal clauses of L . Let $\mathcal{S}_{\oplus}(L)$ denote the set of subformulas of L that are compatible with the operator $\oplus$. For example, for $\oplus = \vee$, the compatible subformulas are disjunctive clauses, while for $\oplus = \wedge$ and $\oplus = \wedge \neg$, the compatible subformulas are conjunctive clauses. For each $S \in \mathcal{S}_{\oplus}(L)$, the algorithm produces a new label

$$L_{\text{ni}} = L[S \leftarrow S \oplus c] \quad (5)$$

where $L[S \leftarrow S \oplus c]$ denotes the formula obtained by replacing the subformula S in L with $S \oplus c$. This operation allows the search to construct formulas that cannot be reached by right-incremental expansions within the same beam tree. For example, inserting O into the internal disjunctive clause $F \vee Q$ in L_1 produces $C_{\text{ni}} = (F \vee Q \vee O) \wedge S$, with induced mask $[1, 0, 1, 1, 0]$. Thus, the incremental and internal expansions of the same label produce different masks and different IoUs with N .

Pairwise recombination candidates. The third family provides a limited form of backtracking by recombining labels selected for expansion at previous beam levels when the resulting label has arity equal to the current level. Given two labels L_1 and L_2 , these combinations are computed using the following rules:

Rule 1: Disjoint recombination. If L_1 and L_2 do not share atomic concepts, then the algorithm generates:

$$L_1 \vee L_2 \quad \text{and} \quad L_1 \wedge L_2 \quad (6)$$

We do not generate $L_1 \wedge \neg L_2$ in this step because compositional explanations assume that negation is applied only to atomic concepts. For example, because in the running example $L_2 = F \vee Q$ and $L_3 = S \wedge O$ have disjoint concept sets, then their recombination produces $C_{\text{rec}_1} = (F \vee Q) \wedge (S \wedge O)$ and $C_{\text{rec}_2} = (F \vee Q) \vee (S \wedge O)$.

Rule 2: Same-type overlapping recombination. Assume that L_1 and L_2 are both clauses formed using the same operator $\oplus_s \in \{\wedge, \vee\}$ and that they share at least one concept. Let $C_{\text{comm}} = C(L_1) \cap C(L_2)$ and $C_{\text{diff}} = C(L_1) \triangle C(L_2)$, where $\triangle$ denotes symmetric difference. Let L_{comm} be the clause obtained by chaining the concepts in C_{comm} using $\oplus_s$. We construct two clauses from C_{diff} : L_{diff}^s , obtained by chaining its concepts using $\oplus_s$, and L_{diff}^c , obtained using the dual operator $\widehat{\oplus}_s$, where $\wedge_c = \vee$ and $\vee_c = \wedge$. The algorithm generates:

$$L_{\text{comm}} \oplus_s L_{\text{diff}}^s \quad \text{and} \quad L_{\text{comm}} \oplus_s L_{\text{diff}}^c \quad (7)$$

For example, in our running example, consider the two disjunctive labels $L_2 = F \vee Q$ and $L_4 = F \vee S$. They share the concept F , while Q and S form the symmetric difference. Rule 2 therefore generates $F \vee (Q \vee S)$ and $F \vee (Q \wedge S)$. The two resulting masks are $[1, 1, 1, 1, 1]$ and $[1, 1, 0, 1, 0]$, respectively.

Rule 3: Specialization of overlapping clauses. Assume that one label is disjunctive and the other is conjunctive, and that they share at least one concept. The algorithm specializes the shared concept in the disjunctive clause by replacing it with the conjunctive clause. For example, considering $L_2 = F \vee Q$ and the conjunctive label $L_5 = F \wedge S$, this rule specializes F in the disjunctive label with $F \wedge S$, producing $(F \wedge S) \vee Q$ which induces the mask $[1, 1, 0, 1, 0]$ in the running example.

Overall, the candidate set at level $k + 1$ is given by:

$$\mathcal{BC}_{k+1} = \mathcal{E}_{\text{inc}}(B_k) \cup \mathcal{E}_{\text{ni}}(B_k) \cup \mathcal{R}_{k+1}(B_{\leq k}) \quad (8)$$

where $\mathcal{E}_{\text{inc}}$ and $\mathcal{E}_{\text{ni}}$ denote incremental and non-incremental expansion, and $\mathcal{R}_{k+1}$ denotes the set of combinations of arity $k + 1$ obtained through pairwise recombination, respectively. The complete pseudo-code of the beam expansion is included in the appendix in Algorithm 1.

3.2.2 Functional-Equivalence-Aware Beam Search

Our proposed algorithm leverages heuristics [21, 20] to guide a beam search algorithm, consistent with prior research [26]. The candidate labels in $\mathcal{BC}_{k+1}$ are ranked according to an admissible heuristic estimate (i.e., upper bound) of their alignment with the neuron activation mask. As a heuristic, we use the one proposed by [20], which derives the upper bound from properties of the atomic concept and activation masks. Specifically, it distinguishes locations that are unique to one concept from those shared by multiple concepts, and whether these locations fall inside or outside the neuron activation mask. It then combines optimistic bounds on these quantities to overestimate the intersection and underestimate the contribution to the union, yielding an admissible upper bound (see Section E for more details about the heuristic). Let $h(L)$ denote the heuristic score associated with a label L , and let us assume $\mathcal{BC}_{k+1}$ is sorted in decreasing order of $h(L)$. The algorithm scans $\mathcal{BC}_{k+1}$ progressively and computes the exact IoU only for candidates whose upper bound is sufficiently high to enter the beam.

Let B_{k+1} denote the current beam at level $k + 1$, with maximum size b . We denote by $L_{\min} = \arg \min_{L \in B_{k+1}} \text{IoU}(N, \circ_L(\mathbf{M}))$ the lowest-scoring label currently in the beam. A candidate L_j is eligible for insertion if $|B_{k+1}| < b$ or $\text{IoU}(N, \circ_{L_j}(\mathbf{M})) \geq \text{IoU}(N, \circ_{L_{\min}}(\mathbf{M}))$. Before inserting L_j , the algorithm checks whether it is functionally equivalent to any label already in the beam. The algorithm distinguishes three cases:

No equivalent label in the beam. In this case L_j is inserted into the beam and, if the beam is full, the lowest-scoring label is removed:

$$B_{k+1} \leftarrow \begin{cases} B_{k+1} \cup \{L_j\} & |B_{k+1}| < b \\ (B_{k+1} \setminus \{L_{\min}\}) \cup \{L_j\} & |B_{k+1}| = b \end{cases} \quad (9)$$

Type-1 and Type-2 equivalence. If L_j is functionally equivalent to a label $L_t \in B_{k+1}$ through Type-1 or Type-2 equivalence, then $C(L_j) = C(L_t)$ and the difference between the two labels is either purely syntactic or induced by dataset-specific redundancy. Retaining both labels would duplicate the same semantic candidate and reduce beam diversity. Therefore, L_j is discarded and L_t is kept, unless L_t is unverified and L_j is verified. In that case, L_t is replaced by L_j :

$$B_{k+1} \leftarrow (B_{k+1} \setminus \{L_t\}) \cup \{L_j\} \quad (10)$$

For example, in the running example, C_{inc} and C_{rec_1} are logically equivalent: $((F \vee Q) \wedge S) \wedge O \equiv (F \vee Q) \wedge (S \wedge O)$. Both therefore induce $[0, 0, 0, 1, 0]$. Once one has entered B_{k+1} , the other is identified as Type-1 functional equivalence and discarded.

Type-3 equivalence. If L_j is functionally equivalent to a label $L_t \in B_{k+1}$ through Type-3 equivalence, the two labels induce the same mask but differ in their concept composition. In this case, the algorithm (i) inserts L_j into the beam, (ii) removes $L_{\min}$ if the beam is full,

$$B_{k+1} \leftarrow \begin{cases} B_{k+1} \cup \{L_j\} & |B_{k+1}| < b \\ (B_{k+1} \setminus \{L_{\min}\}) \cup \{L_j\} & |B_{k+1}| = b \end{cases} \quad (11)$$

and (iii) augments the next candidate sets by recombining L_j and L_t :

$$\mathcal{BC}_t \leftarrow \mathcal{BC}_t \cup \mathcal{R}_t(\{L_j, L_t\}) \quad (12)$$

where t is the length of the formula obtained by the recombination. For example, in the running example, consider $L_t = F \vee S$ and $L_j = Q \vee S$ at the same beam level. Both induce the mask $[1, 1, 1, 1, 1]$, but $C(L_t) = \{F, S\} \neq \{Q, S\} = C(L_j)$, and are therefore Type-3 equivalent. Unlike in Type-1/Type-2 equivalence, L_j is not discarded as redundant. Since the two labels share S , their recombination generates $S \vee (F \vee Q)$ and $S \vee (F \wedge Q)$, which are added as candidates for the next beam level.

The process terminates when either all candidates in BC_{k+1} have been processed or when the next candidate $L_j \in BC_{k+1}$ is associated with an estimate $h(L_j) < \text{IoU}(N, \circ_{L_{\min}}(\mathbf{M}))$ and the beam is full. Since $h(L_j)$ upper-bounds the IoU and the remaining candidates have equal or lower heuristic scores, none of them can obtain an IoU sufficiently high to enter the beam. The current beam is therefore final. The pseudo-code of non-incremental beam search is included in the appendix in Algorithm 2.

4 Experiments

In this section, we evaluate the proposed framework across vision and natural language tasks. We compare our approach against the incremental beam search used in prior work, analyzing both explanation quality and computational cost.

Setup In this paper, we follow the most common setup of compositional explanations. Specifically, we apply our framework across three models trained on Places365 [35] for image classification (ResNet18 [13], AlexNet [18], and DenseNet161 [15]) and three models trained or finetuned on natural language inference tasks (BiLSTM, BERT [7], and TinyLLaMA 1.1B [34]). These models correspond to those investigated in previous literature [26], with the addition of more recent architectures for NLP (see Section H for further details about the models). As probing datasets, we use Broden [3] for image classification, in line with prior work, and SNLI [5] for the language models. For image classification, the concepts correspond to the segmentation masks annotated in the dataset. In the NLI task, we follow [26] and use as the concept set the 2000 most frequent words (tok), part-of-speech tags (tag), overlap between premise and hypothesis, and the neighbors of a word (neighbor), computed as the logical OR between the top 5 closest words in the word2vec space [25]. Following previous approaches, we probe all the neurons in the penultimate layer of each model (the last MLP layer for transformer architectures) and define the neuron activation mask $N \subseteq \Omega$ by selecting the top 0.005 quantile of activations.

Baseline We consider the incremental beam search adopted in previous literature in this research area as a baseline. To the best of the authors’ knowledge, all prior approaches in compositional explanations share the same incremental beam search backbone and therefore converge to the same explanations within the state space. The differences among these approaches mainly lie in the heuristic used to explore the state space [26, 21, 19], the stopping criterion [22], and the concept sets or relations considered [12, 23]. Our approach is orthogonal to these dimensions and can be applied to any of these variants without modification, as they all share the same underlying backbone.

Evaluation Metrics To evaluate our proposal, we consider the following metrics, commonly adopted in this area [26, 22, 4, 21]: IoU, detection accuracy, activation coverage, visited nodes, estimated nodes, and seconds per unit. IoU measures the alignment between the neuron activation mask N and the label mask $\circ_L(\mathbf{M})$. Detection accuracy measures the fraction of concept annotations associated with a label that are recovered within the activation mask N . Activation coverage measures the fraction of neuron activations explained by a label. *Estimated nodes* are candidates for which the heuristic bound $h(L)$ is computed, whereas *visited nodes* are candidates for which the exact IoU is evaluated. For further details on these metrics, we refer to Section B. Finally, we report the value Better (%) as the percentage of neurons for which a beam-based method returns an explanation associated with a strictly higher IoU than the other beam-based method, since the two algorithms explore different beam trees and neither strictly dominates the other.

4.1 Natural Language Inference

In this section, we explore the application of the proposed framework to the natural language inference task. In line with [26], we set the maximum explanation length to 5 and the beam width to 10.

Table 1: Average scores for explanation and search metrics associated with the computation of explanations in the NLP tasks by non-incremental and incremental beam search.

Method	Explanation Quality				Search Metrics		
	Better (%)	IoU	DetAcc	ActCov	Visited	Esti.	s/u
BiLSTM							
Incremental	2	0.171	0.224	0.477	72	370570	37
Ours	58	0.192	0.249	0.483	42	575615	57
BERT							
Incremental	4	0.144	0.189	0.411	54	369531	33
Ours	58	0.156	0.202	0.435	42	726259	62
LLaMA							
Incremental	3	0.179	0.238	0.457	55	364649	34
Ours	65	0.195	0.258	0.471	42	648610	60

Table 1 shows the average scores over all the units (the standard deviation is reported and discussed in Section A). Across all models, our algorithm finds stronger alignments (i.e., higher IoU), and thus better explanations, in more than half of the cases, and produces explanations with higher quality scores across all dimensions. These results confirm the importance, in the NLP domain, of being able to express more complex relationships that go beyond incremental constructions. For example, many neurons in these settings recognize contradictions or coherence between semantics embedded in the premise and hypothesis sentences (e.g., Figure 1). Similarities in semantics often correspond to differences at the syntactic level (e.g., *girl* vs. *woman* vs. *female* for gender). These types of relationships are difficult to express in an incremental formulation, since the scope of the last operator always covers the entire preceding formula, thus limiting the expressiveness of the explanation. Conversely, the possibility of combining candidates and breaking incrementality during beam expansion allows our algorithm to better capture these relationships.

Regarding the search metrics, the additional combinations introduced in the beam search lead to a trade-off in runtime and in the exploration of the search space. However, this trade-off is limited. Specifically, despite the considerably larger number of nodes estimated by the heuristic, the algorithm visits fewer nodes during beam search, and the runtime remains within the same order of magnitude (~ 30 sec vs. ~ 1 min). Summarizing, since in less than 5% of the cases the incremental beam search produces better explanations, our algorithm is the preferred option in these settings.

4.2 Image Classification

In image classification, we consider the settings adopted in prior work [26, 21], fixing the maximum length to 3 and the beam width to 5. Under these settings, it is possible to compute optimal explanations over the state space $\mathcal{L}^n$ in a reasonable time (see Section A.3 for a discussion on the inapplicability of this approach to NLP). Moreover, any explanation retrieved by our proposed algorithm is still contained in $\mathcal{L}^n$, since all non-incremental formulas can be transformed into incremental ones by reordering when the length is less than or equal to 3. In this context, optimal explanations represent the ground truth toward which beam-based explanations aim to converge at a reduced computational cost (~ 2 minutes instead of 1-3 hours). Therefore, we compare the two beam-search approaches both in terms of explanation quality and their ability to recover the optimal solution, and we also report for reference the explanation quality ob-

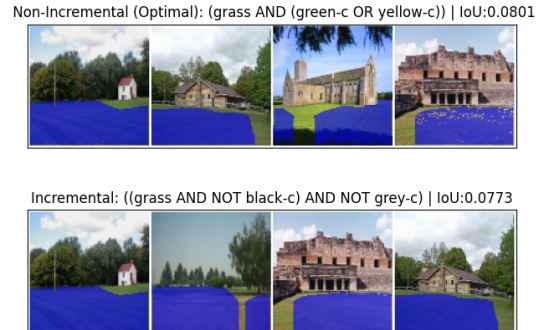


Figure 2: An example of a case where our algorithm reaches optimality while the incremental one does not (Unit 91 of AlexNet). The blue area indicates the activation captured by the explanation.

Table 2: Average scores for explanation and search metrics associated with the computation of explanations in the vision domain by our non-incremental beam search, the standard incremental beam search, and the optimal algorithm.

Method	Explanation Quality					Search Metrics		
	Optimal	Better (%)	IoU	DetAcc	ActCov	Visited	Esti.	s/u
ResNet18								
Incremental	0.92	0	0.073	0.188	0.133	27	33446	122
Ours	0.96	5	0.076	0.181	0.137	71	48154	131
Optimal	1.00	–	0.077	0.176	0.153	–	–	> 3600
AlexNet								
Incremental	0.74	0	0.050	0.155	0.089	61	34054	127
Ours	0.90	16	0.053	0.152	0.089	399	46847	204
Optimal	1.00	–	0.053	0.155	0.090	–	–	> 3600
DenseNet161								
Incremental	0.61	0.33	0.046	0.114	0.087	64	36324	129
Ours	0.76	17	0.048	0.108	0.098	392	49036	180
Optimal	1.00	–	0.048	0.108	0.099	–	–	> 3600

tained by the optimal solutions retrieved using the exact non-beam method [20]. Note that the optimal method maximizes (and is defined for) exclusively the IoU.

Table 2 shows the average scores over all units (the standard deviation is reported in Section A). As expected, the guaranteed-optimal method provides the highest IoU, but at substantially higher computational cost. Our approach significantly increases the number of optimal explanations across all models, representing a substantial step toward closing the gap between fast beam-based explanations and optimal explanations computed at higher cost. In particular, our algorithm is especially effective at reducing the number of non-optimal explanations in cases where beam and optimal explanations achieve similar scores because they share similar concepts but differ in their structure. In these cases, optimal explanations typically return more specific and specialized alignments [20]. Here, the ability to access previous combinations, together with the functional-equivalence-aware beam search, allows the algorithm to correct such cases, avoiding unverified explanations and identifying more specialized combinations of the same concepts (see Figure 2 and Section I).

Regarding runtime, the time required to compute explanations remains close to that of the incremental variant. Although the average is slightly higher, the median time is similar (Section A), suggesting that our algorithm is more sensitive to outliers. Finally, across models, the incremental beam search outperforms our approach in less than 0.5% of cases.

4.3 Human Evaluation

To evaluate whether the additional structural complexity introduced by non-incremental explanations affects their interpretability, we conducted a human study in the NLP setting, where the two formulations can produce substantially different logical structures. We recruited 65 qualified participants, each evaluating six pairs of incremental and non-incremental explanations, for a total of 390 judgments per criterion. In each comparison, participants were shown eight highly activating samples and two anonymized explanations, with their order randomized and balanced. Participants evaluated *understandability*, i.e., which explanation was easier to understand as a description of when the unit activates, and *descriptive power*, i.e., which explanation better described the patterns associated with the unit’s activations. Further study details are reported in Section C.

For understandability, 45.4% of judgments rated the formulations as equivalent, 30.0% preferred the non-incremental explanation, and 24.6% preferred the incremental one. At the participant level, we observed no significant directional preference (two-sided exact sign test, $p = .488$). A non-inferiority analysis with a 10% margin rejects a corresponding loss in understandability for the non-incremental formulation ($p < .001$).

For descriptive power, 42.8% of judgments preferred the non-incremental formulation, 24.9% rated the formulations as equivalent, and 32.3% preferred the incremental one. At the participant level, the preference for the non-incremental formulation was statistically significant (two-sided exact sign test, $p = .018$). Overall, these results provide evidence that the proposed non-incremental structure does not introduce a loss in understandability while improving perceived descriptive power.

5 Analysis and Limitations

This section discusses additional insights and limitations of the proposed algorithm, derived from its design and from further experiments (e.g., ablation studies) reported in Section A.4.

Dependency on heuristics The proposed algorithm relies on an admissible heuristic to guide the search process. As discussed in Section 3, admissibility guarantees that the stopping criterion cannot discard a candidate whose exact IoU could still enter the beam. In this context, the tightness of the heuristic instead affects the computational cost, since a loose but admissible heuristic may substantially increase the number of visited nodes and runtime.

Scalability and Computational Complexity Let p be the number of concepts, b the beam width, n the maximum explanation length, and o the number of operators. Standard incremental beam search generates at most $O(bpo)$ candidates per level, and thus $O(nbpo)$ overall. Our method additionally considers internal expansions and pairwise recombinations over the beam history, resulting in $O(bpo n^2 + b^2 n^2)$ generated candidate formulas in the worst case. Thus, the proposed search has a higher polynomial overhead than incremental beam search, although exploration remains constrained by the beam width. A complete derivation and comparison with guaranteed-optimal search are reported in Section D. In the standard configurations evaluated in this work, this overhead remains moderate, but degenerate cases with many competitive candidates, as reflected by the larger variance between mean and median runtime observed in the vision experiments, can increase runtime.

Memory footprint Functional equivalence requires access to beam masks during the search process. Two strategies can be considered: recomputing masks on demand or caching them in memory. In the first case, the additional memory cost is negligible with respect to the overall setup. In the second case, the memory required to store a single beam is doubled. However, in the setups considered in this paper, we do not observe a significant runtime difference between the two strategies, since functional equivalence checks are relatively infrequent.

6 Conclusion

In this paper, we address the limitations introduced by incremental beam search in compositional explanations. Specifically, we introduced the notion of functional equivalence and proposed a non-incremental beam search framework that expands the beam search space while remaining empirically tractable. We showed that the proposed approach improves the quality of compositional explanations and significantly increases the number of optimal explanations in settings where optimal solutions can be computed. Additionally, we showed that these explanations exhibit higher perceived descriptive power while preserving understandability. These results demonstrate the importance of exploring a broader class of formulas when searching for explanations. Future work could investigate alternative non-incremental search strategies, extend the framework to other logic operators, and explore additional constraints to further improve efficiency and quality of explanations.

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Table 3: Average and standard deviation explanation scores computed by our non-incremental beam search and the standard incremental beam search on natural language inference.

Method	Better (%)	IoU	DetAcc	ActCov
BiLSTM				
Incremental	2	0.171 $\pm$ 0.055	0.224 $\pm$ 0.082	0.477 $\pm$ 0.181
Ours	58	0.192 $\pm$ 0.066	0.249 $\pm$ 0.091	0.483 $\pm$ 0.143
BERT				
Incremental	4	0.144 $\pm$ 0.026	0.189 $\pm$ 0.042	0.411 $\pm$ 0.115
Ours	58	0.156 $\pm$ 0.030	0.202 $\pm$ 0.044	0.435 $\pm$ 0.118
LLaMA				
Incremental	3	0.179 $\pm$ 0.046	0.238 $\pm$ 0.070	0.457 $\pm$ 0.128
Ours	65	0.195 $\pm$ 0.050	0.258 $\pm$ 0.075	0.471 $\pm$ 0.126

Table 4: Average and standard deviation search scores computed by our non-incremental beam search and the standard incremental beam search on natural language inference.

Method	Visited		Estimated	sec/unit	
	Mean	Median		Mean	Median
BiLSTM					
Incremental	72 $\pm$ 115	52	370570 $\pm$ 11364	37 $\pm$ 5	37
Ours	42 $\pm$ 3	41	575615 $\pm$ 176084	57 $\pm$ 15	56
BERT					
Incremental	54 $\pm$ 16	50	369531 $\pm$ 8855	33 $\pm$ 3	33
Ours	42 $\pm$ 3	40	726259 $\pm$ 197093	62 $\pm$ 12	60
LLaMA					
Incremental	55 $\pm$ 16	50	364649 $\pm$ 10608	35 $\pm$ 3	35
Ours	42 $\pm$ 3	40	648610 $\pm$ 186510	60 $\pm$ 12	59

A Complete and Additional Results

A.1 Complete Results

This section reports the standard deviation of the experiments presented in the main text, as well as the median values for the number of visited states and the runtime. Specifically, Table 3 and Table 5 report the scores for the explanation metrics, while Table 4 and Table 6 report the search-related metrics.

It is important to emphasize that the explanations and the metrics used to evaluate their quality are not expected to exhibit low variance. As observed by Bau et al. [3] and Mu and Andreas [26], not all neurons are aligned with specific concepts, leading to high variability in the degree of alignment. This effect is particularly pronounced for metrics computed over pixel-level data, where the alignment between neuron activations and labeled concepts can fluctuate significantly.

In addition to the observations reported in the main text, we note that the median runtime of the proposed algorithm is close to that of incremental beam search in the vision case. This highlights the sensitivity of the algorithm to outliers, where outliers correspond to neurons for which multiple candidate explanations achieve similar IoU scores (e.g., neurons with overall low alignment). At the same time, this indicates that, for a large portion of cases, the proposed algorithm runs in a time comparable to the incremental version in the vision case, which is a desirable property.

A.2 Heuristics Comparison

For completeness, we compare in Table 7 the original MMESH heuristic [21] in the same vision settings analyzed in this paper with respect to the heuristic adopted [20] in this paper. Under the same incremental beam-search formulation, the heuristic adopted in this work requires fewer exact

Table 5: Average and standard deviation explanation scores computed by our non-incremental beam search and the standard incremental beam search on image classification.

Method	Optimal	Better (%)	IoU	DetAcc	ActCov
ResNet18					
Incremental	0.92	0	0.073 $\pm$ 0.042	0.188 $\pm$ 0.108	0.133 $\pm$ 0.090
Ours	0.96	5	0.076 $\pm$ 0.044	0.181 $\pm$ 0.106	0.137 $\pm$ 0.088
AlexNet					
Incremental	0.74	0	0.050 $\pm$ 0.024	0.155 $\pm$ 0.101	0.089 $\pm$ 0.066
Ours	0.90	16	0.053 $\pm$ 0.024	0.152 $\pm$ 0.093	0.089 $\pm$ 0.057
DenseNet161					
Incremental	0.61	0.33	0.037 $\pm$ 0.022	0.094 $\pm$ 0.058	0.078 $\pm$ 0.072
Ours	0.76	17	0.038 $\pm$ 0.022	0.083 $\pm$ 0.073	0.081 $\pm$ 0.052

Table 6: Average and standard deviation of search scores computed by our non-incremental beam search and the standard incremental beam search on image classification.

Method	Visited		Estimated	s/u	
	Mean	Median		Mean	Median
ResNet18					
Incremental	27 $\pm$ 40	14	33446 $\pm$ 2628	122 $\pm$ 12	117
Ours	71 $\pm$ 204	17	48154 $\pm$ 5721	131 $\pm$ 34	122
AlexNet					
Incremental	61 $\pm$ 48	40	34054 $\pm$ 2390	127 $\pm$ 8	126
Ours	399 $\pm$ 624	102	46847 $\pm$ 7639	204 $\pm$ 134	137
DenseNet161					
Incremental	64 $\pm$ 47	60	36324 $\pm$ 2812	129 $\pm$ 12	126
Ours	392 $\pm$ 742	104	49036 $\pm$ 8169	180 $\pm$ 125	129

evaluations and lower runtime than the original MMESH heuristic across all three vision models. These results are in line with and confirm the results provided in [20], where the reader can find a more extensive comparison between the two heuristics.

A.3 Optimal Explanations for NLP

As discussed in the main text, in the vision setting and for explanations of maximum length 3, it is possible to compute ground-truth optimal explanations in a reasonable time using the algorithm proposed by La Rosa and Gilpin [20]. This algorithm requires between 1 and 3 hours per neuron and assumes an incremental formulation. As noted in their work, the algorithm relies on the presence of *unique elements* in the dataset, defined as elements associated with a single annotation, and its performance is sensitive to the length of the explanation.

We attempted to apply this algorithm to the NLP setting considered in this paper. However, in this case, the required assumptions do not hold. In particular, there are no unique elements, since each word is associated with at least two concepts (e.g., token and part-of-speech tag). Moreover, since activations are computed at the sentence level, many words are treated as *common* elements despite being distinct. Given these properties, together with the larger maximum explanation length considered in NLP, we observe that the optimal algorithm does not run in a feasible time in this setting. This limitation further motivates the need for approximate or search-based methods in settings where optimal computation is infeasible.

Finally, note that while any explanation of length 3 can be reordered into an incremental form and thus compared with the optimal solution, this is not the case for explanations of length greater than 3. Therefore, even if the optimal algorithm were applied to longer explanations in the vision setting, the

Table 7: Search metrics for incremental beam search using the heuristic adopted in this work and the MMESH heuristic.

Model	Heuristic	Visited	Estimated	s/u
ResNet18	Adopted	27	33,446	122
	MMESH	48	53,752	140
AlexNet	Adopted	61	34,054	127
	MMESH	94	53,753	174
DenseNet161	Adopted	64	36,324	129
	MMESH	140	53,753	196

Table 8: Average scores and standard deviation for explanation metrics when individual components of our framework are removed.

Method	Better (%)	IoU	DetAcc	ActCov
w/o Recombinations	0.64	0.145 $\pm$ 0.021	0.183 $\pm$ 0.030	0.442 $\pm$ 0.121
Full	3.18	0.151 $\pm$ 0.026	0.197 $\pm$ 0.027	0.417 $\pm$ 0.124
w/o Functional Equivalence	4	0.135 $\pm$ 0.016	0.174 $\pm$ 0.027	0.397 $\pm$ 0.086
Full	60	0.147 $\pm$ 0.021	0.187 $\pm$ 0.041	0.445 $\pm$ 0.109
w/o Non-Incremental Expansion	4	0.137 $\pm$ 0.017	0.182 $\pm$ 0.033	0.382 $\pm$ 0.092
Full	66	0.147 $\pm$ 0.020	0.190 $\pm$ 0.042	0.435 $\pm$ 0.111

results would not be directly comparable. In this direction, future work should explore the notion of optimality in non-incremental settings.

A.4 Ablation Study and Sensitivity

Here, we analyze the proposed algorithm under several configurations. As the probed model, we use BERT on the natural language inference task.

A.4.1 Ablation Study

Table 8 shows the difference in performance when ablating individual components of the proposed approach. In particular, we consider four configurations: *full*, which includes all components; *w/o recombination*, which disables the generation of pairwise recombination candidates during beam expansion as well as the combinations induced by functional equivalence; *w/o Functional Equivalence*, which removes the checks for functional equivalence among promising candidates; and *w/o Non-Incremental Expansion*, which removes non-incremental candidates from the beam expansion while retaining recombinations.

Similarly to the main text, we generate explanations for both the ablated and full configurations and compare the explanations that differ pairwise. From Table 8, we observe that functional equivalence and non-incremental expansion are the most impactful components, as the full version finds better alignments in more than 60% of the cases than the ablated versions without these specific components and achieves higher scores across all metrics. Conversely, recombinations have a smaller impact but still contribute to improved overall performance.

A.4.2 Sensitivity to Beam Size and Explanation Length

In this section, we study the sensitivity of the proposed algorithm to variations in the hyperparameters related to beam width and explanation length. In practice, both parameters are usually kept relatively small to preserve interpretability and tractable runtime. However, due to the inclusion of recombinations and the larger search space explored by the proposed method, we investigate whether the runtime gap between the incremental and non-incremental variants grows linearly or exponentially with respect to these parameters.

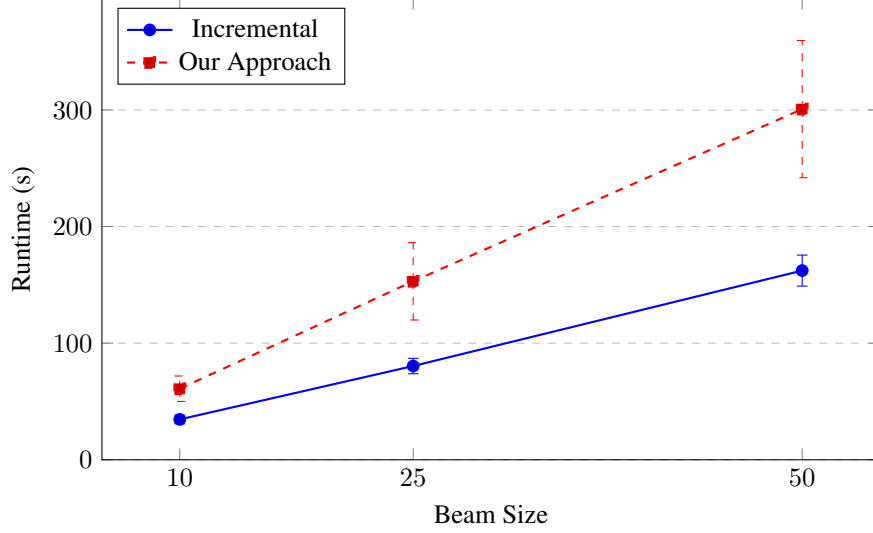


Figure 3: Runtime vs Beam Size. Error bars denote standard deviation.

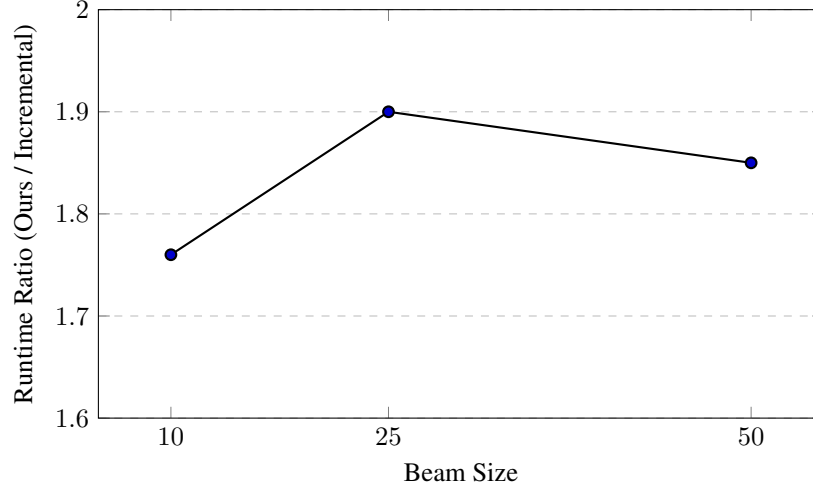


Figure 4: Runtime ratio between our approach and incremental beam search across beam sizes.

Figures 4 and 6 show that, for both beam width and explanation length, the ratio between the runtime of the incremental and non-incremental variants remains approximately constant, indicating similar scaling behavior.

However, considering the absolute values reported in Figures 3 and 5, it is clear that, in extreme configurations involving very large beam widths and long explanations, the proposed algorithm may become impractical to apply. Nevertheless, we argue that these settings are rare in practice. Indeed, it has already been observed that, for complex datasets such as Broden, beam widths larger than 20 become impractical even for incremental beam search, due to the combination of large mask sizes and the high number of closely related formulas in the search space. Still, we believe this limitation is important to highlight.

A natural follow-up question is **whether the gains in explanation quality are simply due to the relatively small beam sizes** typically adopted in prior work, and whether the incremental formulation could recover similar explanations by increasing the beam width. Our experiments suggest that the answer is negative. Table 9 reports a comparison of the explanation metrics when the beam width is fixed to 50 on 50 randomly extracted units. We observe that the gap between the incremental and non-incremental approaches remains consistent with the one observed in the main text, highlighting

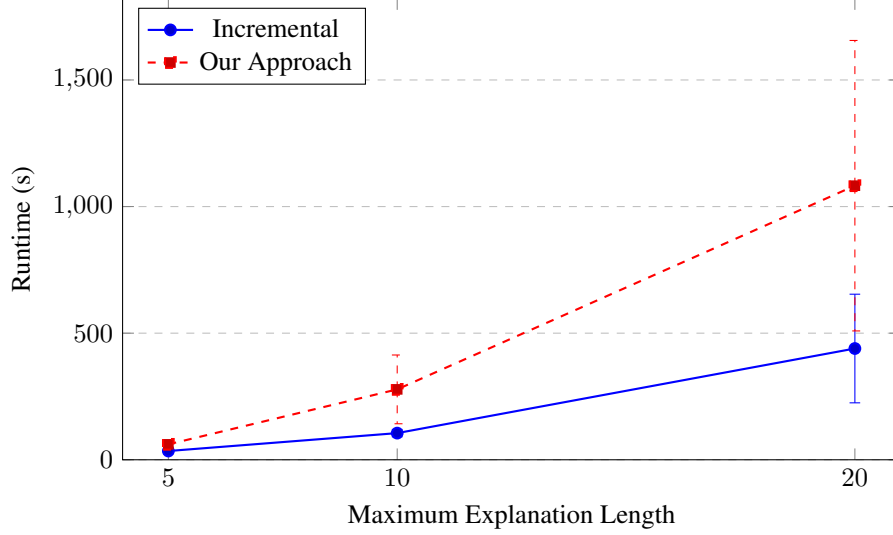


Figure 5: Runtime as a function of the maximum explanation length. Error bars denote standard deviation.

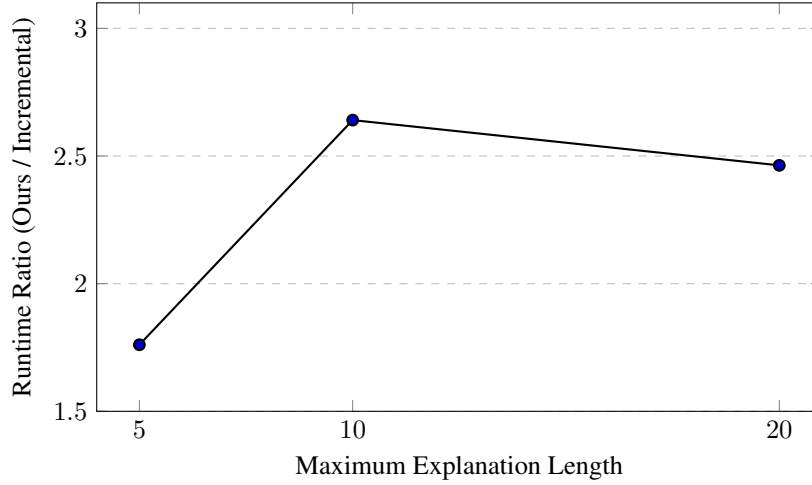


Figure 6: Runtime ratio between our approach and incremental beam search as a function of the maximum explanation length.

that the improvements are not tied to a specific beam configuration. These results suggest that the gains are primarily due to the expanded search space enabled by the non-incremental formulation, rather than to differences in beam width alone.

B Metrics

This section reports the Detection Accuracy and Activation Coverage definitions, as introduced in [21, 22]. In the following definitions, $N \subseteq \Omega$ denotes the neuron activation mask and $\circ_L(\mathbf{M}) \subseteq \Omega$ denotes the mask induced by label L over the probing dataset.

Detection Accuracy [22] quantifies the fraction of the label mask that is captured by the neuron activation mask. A high value indicates that most of the regions associated with the label are detected by the neuron within the considered activation range.

$$DetAcc(L, N, \mathbf{M}) = \frac{|N \cap \circ_L(\mathbf{M})|}{|\circ_L(\mathbf{M})|} \quad (13)$$

Table 9: Average and standard deviation explanation scores computed by our non-incremental beam search and the standard incremental beam search on BERT using a beam width equal to 50.

Method	Better (%)	IoU	DetAcc	ActCov
Incremental	0	0.144 $\pm$ 0.026	0.190 $\pm$ 0.044	0.407 $\pm$ 0.109
Ours	69	0.154 $\pm$ 0.022	0.198 $\pm$ 0.037	0.434 $\pm$ 0.096

Activation Coverage [21] measures the fraction of neuron activations that fall within the regions associated with the label. A high value indicates that the label covers a large portion of the activation mask.

$$ActCov(L, N, \mathbf{M}) = \frac{|N \cap \circ_L(\mathbf{M})|}{|N|} \quad (14)$$

C User Study

To evaluate whether the additional structural complexity introduced by non-incremental explanations affects their interpretability, we conducted a human study in the NLP setting, where the two formulations can produce different logical structures.

To conduct this study, we recruited participants through Prolific. Participants were required to have English as their primary language, an approval rate greater than 90%, and more than 100 previous submissions. We additionally required either Prolific’s qualified AI tasker status or a background in computer science. The study was pre-approved under an exemption by the University of California, Santa Cruz Institutional Review Board and is currently undergoing the full approval process. Informed consent was obtained from all participants. Participation was voluntary, and participants could withdraw before submitting the survey. Participants received \$5 for completing the study, corresponding to an average compensation of \$9.42 per hour. The final analysis included 65 qualified participants. Before the evaluation, participants completed a tutorial explaining the logical operators, parentheses, and the two evaluation criteria, followed by five qualification questions. We retained participants who answered at least four of the five qualification questions correctly.

Each qualified participant evaluated six comparisons between an incremental and a non-incremental explanation, resulting in 390 judgments per criterion. For each comparison, participants were shown eight premise–hypothesis pairs associated with a highly activating unit together with two explanations, anonymously presented as Explanation A and Explanation B. The assignment of the incremental and non-incremental explanations to A and B was randomized and balanced, and both explanations were evaluated using the same set of examples. Participants evaluated two criteria. *Descriptive power* asked: “Considering all eight sentence pairs, which explanation better describes the pattern or patterns associated with this unit’s activation?” *Understandability* asked: “Which explanation was easier to understand as a description of when this unit activates?” For each criterion, participants selected one of five ordered responses: Explanation A is much better, Explanation A is better, the explanations are about equally good, Explanation B is better, or Explanation B is much better.

For the aggregate analysis, responses were grouped into three categories: preference for the incremental explanation, equivalence, and preference for the non-incremental explanation. Since each participant provided multiple judgments, statistical tests were performed at the participant level rather than treating the 390 judgments as independent. For each criterion, we counted whether each participant favored the non-incremental formulation more often, favored the incremental formulation more often, or was tied. Directional preference was evaluated using a two-sided exact sign test, excluding ties. For understandability, we additionally performed a non-inferiority analysis. For each participant, we computed the difference between the proportion of judgments favoring the non-incremental and incremental formulations and tested non-inferiority using a 10% margin.

Results. For descriptive power, 42.8% of the judgments favored the non-incremental explanation, 24.9% considered the explanations equivalent, and 32.3% favored the incremental explanation. At the participant level, 35 participants favored the non-incremental formulation more often, 17 favored the

incremental formulation more often, and 13 were tied. This corresponds to a statistically significant preference for the non-incremental formulation (two-sided exact sign test, $p = .018$).

For understandability, 30.0% of the judgments favored the non-incremental explanation, 45.4% considered the explanations equivalent, and 24.6% favored the incremental explanation. At the participant level, 29 participants favored the non-incremental formulation more often, 23 favored the incremental formulation more often, and 13 were tied. We did not observe a significant directional preference (two-sided exact sign test, $p = .488$). Using a 10% non-inferiority margin, we reject that the non-incremental formulation introduces a corresponding loss in understandability ($p < .001$).

These results suggest that the structural complexity introduced by non-incremental explanations can improve how well explanations characterize neuron behavior without making them harder to understand in the setting considered in this paper.

D Complexity Analysis

In this section, we analyze the worst-case number of candidate formulas generated by each search procedure. This quantity is the major factor influencing the running time because generated candidates require a heuristic estimate and may require an exact IoU computation, which is the most expensive operation. We characterize this quantity using the number of atomic concepts $p = |\mathcal{L}^1|$, the beam width b , the maximum explanation length n , and the number of admissible operators $o = |\mathcal{O}|$.

Incremental beam search. At each level, the beam contains at most b labels. Each label can be expanded using at most p concepts and o operators. Therefore, the incremental formulation generates at most $O(bpo)$ candidates per level. Therefore, across at most n levels, incremental beam search generates $O(nbpo)$ candidate formulas.

Non-incremental beam search. Our formulation retains the incremental expansions and additionally generates candidates through internal insertion and pairwise recombination.

For internal insertion, a formula of arity k contains at most $k - 1$ internal operator positions at which a new concept can be inserted. Since at most b formulas are expanded using at most p concepts and o operators, the number of internal-expansion operations at level k is bounded by $O(kbpo)$. Summing over all levels results in $\sum_{k=1}^n O(kbpo) = O(bpo n^2)$ candidate formulas.

For pairwise recombination, pairs considered at previous levels are not recomputed. When the beam B_k is generated, the algorithm considers pairs between formulas in B_k and formulas retained at preceding levels, as well as pairs within B_k . The current beam contains at most b formulas, while the search history contains at most bk formulas. Therefore, the number of new pairs considered at level k is at most $O(b^2k)$. Summing over all levels, the total number of distinct pairwise recombination attempts is $\sum_{k=1}^n O(b^2k) = O(b^2n^2)$.

Including incremental, internal expansions, and pairwise recombination, the total number of candidate formulas generated by our method is therefore

$$O(nbpo + bpo n^2 + b^2n^2) = O(bpo n^2 + b^2n^2)$$

Optimal search. The optimal method [20] does not constrain the search to a fixed-width beam. Therefore, in the worst case, it may need to explore the full admissible state space. Under the assumption that each concept appears at most once, the number of incremental formulas up to length n is upper-bounded by

$$\sum_{k=1}^n o^{k-1} (p)_k$$

where $(p)_k = \frac{p!}{(p-k)!}$ is the number of ordered selections of k distinct concepts. For fixed n , this quantity is polynomial in p , but it grows combinatorially as the maximum explanation length increases.

The admissible heuristic used by the guaranteed-optimal method can substantially reduce the number of explored states in practice, but in the worst case its behavior approaches exhaustive exploration.

Our method trades the optimality guarantee for a beam search whose computational cost of candidate generation is polynomial in p , b , o , and n , while empirically recovering optimal explanations more frequently than standard incremental beam search.

Additional computational operations. The bounds above count generated candidate formulas and abstract from the elementary operations required to construct them. These costs can introduce additional dependence on formula length. For example, processing a recombination pair may require $O(k)$ operations in the worst case to inspect the concepts in formulas of length at most k , identify shared and non-shared concepts, and construct the resulting formula. Consequently, although only $O(b^2n^2)$ distinct recombination pairs are considered across the search, their structural processing can require up to $O(b^2n^3)$ operations in the worst case. Similarly, identifying compatible subformulas and constructing internally modified formulas can introduce additional formula-length-dependent operations. However, these structural operations are order-of-magnitude faster than mask-based computations used during search. For this reason, we excluded them from the complexity asymptotic analysis above, and we report wall-clock runtime in the experiments to capture the remaining implementation costs.

E Heuristics

Our algorithm uses the admissible heuristic introduced in [20] for estimating the IoU of candidate formulas. Specifically, we use the sample-based label-quantity estimation. We use this heuristic since it is compatible with both vision and NLP settings and does not assume the incremental formulation.

The heuristic decomposes the locations in Ω according to whether they are associated with a single atomic concept (unique elements U) or multiple concepts (common elements C). We define

$$U = \left\{ \omega \in \Omega : \sum_{c \in \mathcal{L}^1} \mathbb{1}[\omega \in M_c] = 1 \right\}$$

$$C = \left\{ \omega \in \Omega : \sum_{c \in \mathcal{L}^1} \mathbb{1}[\omega \in M_c] > 1 \right\}$$

Starting from these definitions, the neuron activations are partitioned into $N^U = N \cap U$ and $N^C = N \cap C$, respectively. We also define the space available for extra locations as:

$$SE^U = U \setminus N$$

$$SE^C = C \setminus N$$

Given a candidate formula L , we define its unique and common intersection locations as:

$$I^U(L) = \circ_L(\mathcal{M}) \cap N^U$$

$$I^C(L) = \circ_L(\mathcal{M}) \cap N^C$$

and its unique and common extra locations as

$$E^U(L) = \circ_L(\mathcal{M}) \cap SE^U,$$

$$E^C(L) = \circ_L(\mathcal{M}) \cap SE^C.$$

Using these quantities, the IoU can be equivalently expressed as:

$$\text{IoU}(N, \circ_L(\mathcal{M})) = \frac{|I^U(L)| + |I^C(L)|}{|N| + |E^U(L)| + |E^C(L)|}$$

Because unique locations cannot be shared by distinct concepts, their contribution can be computed exactly from the operators in the formula. Conversely, masks of different concepts may overlap in

common locations. Following [20], we precompute whether concept masks are disjoint. If they are disjoint, the corresponding quantities can be computed exactly. Otherwise, the heuristic estimates an upper bound on the common intersection ($I_{\max}^C$) and a lower bound on the common extras ($E_{\min}^C$).

In the following, $L_{\leftarrow}$ denotes the partial formula already retained by beam search, while $L_{\rightarrow}$ denotes the atomic concept used to expand it. When a formula enters the beam, its quantities are computed and retained together with its IoU. Therefore, the quantities associated with $L_{\leftarrow}$ are already available. Similarly, because $L_{\rightarrow}$ is an atomic concept, its quantities can be precomputed from the probing dataset. The heuristic therefore only needs to estimate how these quantities combine under the operator $\oplus$, and in particular the unknown overlap within the common locations.

For a formula $L = L_{\leftarrow} \oplus L_{\rightarrow}$, these bounds are computed per sample. For OR:

$$\begin{aligned} |I_{\max}^C(L)_x| &= \min(|I_{\max}^C(L_{\leftarrow})_x| + |I^C(L_{\rightarrow})_x|, |N_x^C|) \\ |E_{\min}^C(L)_x| &= \max(|E_{\min}^C(L_{\leftarrow})_x|, |E^C(L_{\rightarrow})_x|) \end{aligned}$$

For AND:

$$\begin{aligned} |I_{\max}^C(L)_x| &= \min(|I_{\max}^C(L_{\leftarrow})_x|, |I^C(L_{\rightarrow})_x|) \\ |E_{\min}^C(L)_x| &= \max(|E_{\min}^C(L_{\leftarrow})_x| + |E^C(L_{\rightarrow})_x| - |SE_x^C|, 0) \end{aligned}$$

Finally, for AND NOT:

$$\begin{aligned} |I_{\max}^C(L)_x| &= \min(|I_{\max}^C(L_{\leftarrow})_x|, |N_x^C| - |I^C(L_{\rightarrow})_x|) \\ |E_{\min}^C(L)_x| &= \max(|E_{\min}^C(L_{\leftarrow})_x| - |E^C(L_{\rightarrow})_x|, 0) \end{aligned}$$

These estimates correspond to the most favorable overlap compatible with the information available for the two operands. The heuristic IoU is then computed as:

$$h(L) = \frac{\sum_{x \in \mathcal{D}} (|I^U(L)_x| + |I_{\max}^C(L)_x|)}{|N| + \sum_{x \in \mathcal{D}} (|E^U(L)_x| + |E_{\min}^C(L)_x|)}$$

The heuristic is admissible because the unique quantities are exact and the estimated common quantities satisfy $|I_{\max}^C(L)_x| \geq |I^C(L)_x|$ and $|E_{\min}^C(L)_x| \leq |E^C(L)_x|$. Thus, the heuristic cannot underestimate the IoU numerator or overestimate its denominator, and therefore $h(L) \geq \text{IoU}(N, \circ_L(\mathcal{M}))$. A complete derivation and proof of admissibility for the considered operators are provided in [20].

This admissibility property justifies the stopping criterion used in our beam selection. Since candidates are processed in decreasing order of $h(L)$, once the next candidate has a heuristic value lower than the exact IoU of the lowest-scoring formula in a full beam, neither that candidate nor any subsequent candidate can enter the beam.

F Algorithm Design Choices

This section motivates and further expands on some of the design choices we took in proposing the non-incremental algorithm. Overall, our choices and rules should be viewed as a conservative first design rather than as an optimal trade-off between expressivity and computation.

Expansion Rules Motivations. The proposed expansion rules represent a first attempt to support non-incremental search strategies. We designed the proposed rules to enlarge the reachable search space while keeping candidate generation manageable and preserving the assumption that each concept appears at most once in an explanation. The incremental expansion preserves the standard search paths considered in prior work. The internal insertion addresses a failure mode observed when a concept that improves the explanation may need to be inserted within an existing subformula rather than appended to the complete formula [20]. Pairwise recombination instead provides a limited form of backtracking, allowing the algorithm to combine structures retained at different stages of the search rather than extending only the current beam.

Recombinations When combining formulas from the beam history, formulas that do not share concepts provide the simplest case because no duplicated concepts need to be resolved. When the formulas share concepts, we restrict recombination to the structures considered in Rules 2 and 3, where duplicated concepts can be removed without requiring general logical simplification. More general recombinations between arbitrary overlapping formulas could introduce repeated concepts or require transformations to preserve the semantics of the original formulas. We leave these broader recombination strategies to future work.

Recombinations beyond Beam Candidates. In preliminary experiments, we also considered the configuration in which recombination used the top- q candidates at each level, with $q > b$, rather than only formulas retained in the beam. These variants produced some additional improvements in alignment, but increased both runtime and the amount of search history that needed to be maintained. They would also combine the effect of non-incremental recombination with that of retaining a larger candidate pool than standard beam search. We therefore restrict recombination to beam candidates in the formulation evaluated in this work.

G Broader Impact Statement

The contributions of this work are mainly experimental; they are not deployed in downstream applications and refer mainly to algorithmic aspects. Moreover, this work does not alter sensitive components of the explanation pipeline. As such, we believe that this work doesn't improve or degrade the broader impact of explainability methods in this area of research.

H Reproducibility

This section serves as a brief summary and documentation of the experimental setup used by our framework, along with the resources required.

H.1 Dataset, Models, and Explanations

In this section, we provide the repository, dataset, explanations, and model information, versions, their corresponding licenses, download links, and a brief description of the modifications required to ensure compatibility with our framework.

Probing Datasets

- Broden
 - Accessible at: <https://github.com/CSAILVision/NetDissect>
 - License: MIT
- SNLI v.1.0
 - Accessible at: <https://nlp.stanford.edu/projects/snli/>
 - License: CC-BY-SA

Probed Models

- AlexNet
 - Accessible at: http://places2.csail.mit.edu/models_places365/alexnet_places365.pth.tar
 - License: MIT
- ResNet18
 - Accessible at: http://places2.csail.mit.edu/models_places365/resnet18_places365.pth.tar
 - License: MIT

- DenseNet161
 - Accessible at: http://places2.csail.mit.edu/models_places365/whole_densenet161_places365_python36.pth.tar
 - License: MIT
- BiLSTM
 - Accessible at: http://downloads.cs.stanford.edu/nlp/data/muj/bowman_snli/6.pth
 - License: MIT

To these models, we add BERT and LLaMA. The pretrained models are respectively accessible at <https://huggingface.co/google-bert/bert-base-uncased> and <https://huggingface.co/knowledgator/Llama-encoder-1.0B>, both associated with Apache License 2.0. For both, we follow the structure proposed in [26]. Specifically, we replace the LSTM encoder with a BERT encoder and a LLaMA-based encoder, respectively, and adjust the input dimension of the MLP according to the hidden size of the encoder. We train BERT and LLaMA using the AdamW optimizer for 5 epochs with a learning rate of $2e-5$, following the training setup included in <https://github.com/jayelm/compexp>. All models converge to an accuracy greater than 80%, which matches the performance of the BiLSTM baseline.

Explanations Our framework generates explanations through a heuristic search guided by the heuristic proposed in [20]. We reimplemented the heuristic and applied it to the NLP domain following the pipeline available at <https://github.com/jayelm/compexp>. Regarding the hyperparameters, we follow the setup proposed in [26], analyzing the top 0.005 quantile of activations, setting the explanation length to 3 (vision) and 5 (NLP), setting the beam size to 5 (vision) and 10 (NLP), and employing the AND, OR, AND NOT operators, and considering the same concept set defined in [26].

H.2 Resources

The proposed algorithm introduces additional computational and memory overhead with respect to incremental compositional explanations, as discussed in Section D. In terms of memory, two strategies can be considered: caching the beam masks for functional equivalence checks or recomputing them on demand. When masks are recomputed on demand, the additional memory cost introduced by our modifications is negligible with respect to the overall setup. When masks are cached, the memory footprint increases, approximately doubling the memory required to store a single beam. Note that, in the experiments reported in this paper, we do not observe a large difference in runtime between the two strategies. However, in general settings, recomputing the masks may become a significant bottleneck.

All experiments were conducted on a workstation equipped with an NVIDIA RTX 3090 GPU, 8 CPU cores, and 64 GB of RAM. The runtimes reported in the main text and in this appendix are based on this setup. However, the experiments can be replicated on significantly lighter configurations.

The overall computational cost depends on (i) the data (i.e., domain), (ii) the number of concepts, and (iii) the probed model. These factors are independent of the proposed method. The data type influences the representation of concept masks, neuron activations, and the information required by the heuristic. In vision settings, these are typically matrices, resulting in a heavier memory footprint (~ 10 GB). In the NLP setting analyzed in this paper, they are vectors, leading to a significantly lighter setup (~ 1 GB). The number of concepts is also related to the data domain: a larger concept set increases memory usage. As a reference, the ~ 10 GB estimate corresponds to a concept dataset with approximately 1200 concepts in the vision domain.

Finally, the probed model impacts the computation only in the preprocessing stage. The model is used to compute neuron activations but is not required during the explanation phase. Therefore, once the activations are computed, the model can be discarded. In this context, the required memory depends on the specific model, ranging from a few MB for the BiLSTM (Bowman) model to approximately ~ 20 GB for LLaMA.

I Visualization of Explanation Differences

This section reports examples of explanations computed by incremental and non-incremental beam search. Specifically, we focus on cases where the explanations differ. Note that **compositional explanations are computed over a much larger set of samples than the one visualized**. The visualizations are provided only to contextualize the highlighted differences between explanations and should not be interpreted as a direct comparison method. Therefore, the images shown do not represent the only and specific samples explained by the labels, but they represent a limited representative subset of the state space over which the explanations are computed.

I.1 Vision

For vision, we follow the setup used in [21]. Specifically, for each visualization, we select the samples (i) associated with the highest intersection between the explanation area and the activation area (to ease interpretation), (ii) where the explanation holds, and (iii) where the neuron is active within the considered activation range. The examples (Figures 7 to 12) are not cherry-picked and correspond to the first six differing explanations for the last convolutional layer of each explained model (ResNet18 [13], AlexNet [18], and DenseNet [15]). We indicate with a check mark whether one of the two explanations is optimal. Note that the *-s* included in some of the concepts refer to scene concepts, which extend over the whole image.

I.2 Natural Language Inference

For NLP, we follow a setup similar to the one adopted for the vision case. Specifically, for each visualization, we select the samples (i) associated with the highest activations, (ii) for which the explanation holds, and (iii) for which the neuron is active within the considered activation range. Note that this visualization differs from the one used in Figure 1, since we filter out samples for which the explanation is not active. This choice is motivated by the same considerations discussed in the vision setting: simply visualizing the top-*k* activating samples would often result in examples where none of the explanations hold, since explanations are computed over a much larger sample set and visualizing all of them is impractical.

As in the vision setting, the examples (Figures 13 to 29) are not cherry-picked and correspond to the first six units where the explanations differ for the MLP layer of each analyzed model (BiLSTM [26], BERT [7], and LLaMA [31]).



Figure 7: Alignment detected incremental and non-incremental beam-search methods on ResNet18. The blue regions indicate areas of neuron activation within the considered range. The checkmark indicates whether one of two explanations is optimal.

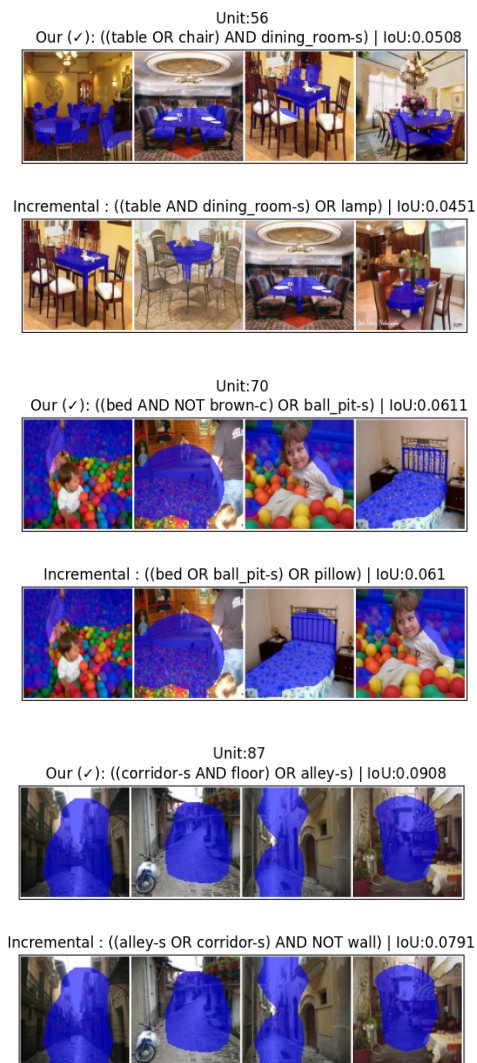


Figure 8: Alignment detected incremental and non-incremental beam-search methods on ResNet18. The blue regions indicate areas of neuron activation within the considered range. The checkmark indicates whether one of two explanations is optimal.



Figure 9: Alignment detected incremental and non-incremental beam-search methods on AlexNet. The blue regions indicate areas of neuron activation within the considered range. The checkmark indicates whether one of two explanations is optimal.



Figure 10: Alignment detected incremental and non-incremental beam-search methods on AlexNet. The blue regions indicate areas of neuron activation within the considered range. The checkmark indicates whether one of two explanations is optimal.



Figure 11: Alignment detected incremental and non-incremental beam-search methods on DenseNet161. The blue regions indicate areas of neuron activation within the considered range. The checkmark indicates whether one of two explanations is optimal.

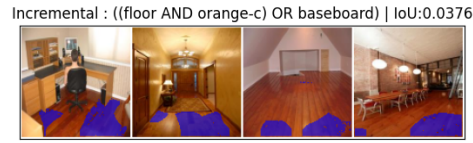
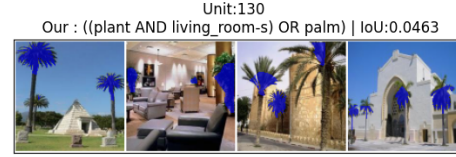


Figure 12: Alignment detected incremental and non-incremental beam-search methods on DenseNet161. The blue regions indicate areas of neuron activation within the considered range. The checkmark indicates whether one of two explanations is optimal.

Unit 0 – Incremental Beam Search IoU: 0.136 Explanation $(((\text{hyp:tok:alone} \vee \text{pre:tok:sleeves}) \wedge \neg \text{hyp:tag:rb}) \vee \text{hyp:tok:sleeps}) \wedge \text{neighbor:pre:tok:with})$ Samples selected by this explanation <table> <tr> <th>Premise / Hypothesis</th> <th>Ours Applies</th> </tr> <tr> <td>P: On a sunny day, in a lush, green field, one small girl inspects a rock while her friend goes off in another direction. H: A girl sleeps</td> <td>●</td> </tr> <tr> <td>P: A woman dressed in a blue long-sleeved shirt and wearing a hairnet. H: The woman is naked and alone in the bathroom.</td> <td>●</td> </tr> <tr> <td>P: A girl in a purple sweater and a shoulder bag is looking to the side with a lot of people behind her. H: A boy sleeps on the couch.</td> <td>●</td> </tr> <tr> <td>P: A man with a black shirt holds a baby while a blue shirted individual faces them. H: A man sits all alone on a park bench, arms folded.</td> <td>●</td> </tr> <tr> <td>P: Two women dressed for cold weather in jackets and gloves look at something on a cellphone screen. H: The woman is alone in her bathroom.</td> <td>●</td> </tr> </table>	Premise / Hypothesis	Ours Applies	P: On a sunny day, in a lush, green field, one small girl inspects a rock while her friend goes off in another direction. H: A girl sleeps	●	P: A woman dressed in a blue long-sleeved shirt and wearing a hairnet. H: The woman is naked and alone in the bathroom.	●	P: A girl in a purple sweater and a shoulder bag is looking to the side with a lot of people behind her. H: A boy sleeps on the couch.	●	P: A man with a black shirt holds a baby while a blue shirted individual faces them. H: A man sits all alone on a park bench, arms folded.	●	P: Two women dressed for cold weather in jackets and gloves look at something on a cellphone screen. H: The woman is alone in her bathroom.	●	Unit 0 – Our Non-Incremental Search IoU: 0.149 Explanation $(((\text{pre:tok:sleeves} \vee \text{hyp:tok:alone}) \vee \text{hyp:tok:sleeps}) \wedge (\text{pre:tag:vbn} \vee \text{neighbor:pre:tok:individual}))$ Samples selected by this explanation <table> <tr> <th>Premise / Hypothesis</th> <th>Inc. Applies</th> </tr> <tr> <td>P: A woman dressed in a blue long-sleeved shirt and wearing a hairnet. H: The woman is naked and alone in the bathroom.</td> <td>●</td> </tr> <tr> <td>P: A band is performing on stage for people. H: A band practicing alone in their parents' basement.</td> <td>●</td> </tr> <tr> <td>P: A girl in a purple sweater and a shoulder bag is looking to the side with a lot of people behind her. H: A boy sleeps on the couch.</td> <td>●</td> </tr> <tr> <td>P: A man with a black shirt holds a baby while a blue shirted individual faces them. H: A man sits all alone on a park bench, arms folded.</td> <td>●</td> </tr> <tr> <td>P: Two women dressed for cold weather in jackets and gloves look at something on a cellphone screen. H: The woman is alone in her bathroom.</td> <td>●</td> </tr> </table>	Premise / Hypothesis	Inc. Applies	P: A woman dressed in a blue long-sleeved shirt and wearing a hairnet. H: The woman is naked and alone in the bathroom.	●	P: A band is performing on stage for people. H: A band practicing alone in their parents' basement.	●	P: A girl in a purple sweater and a shoulder bag is looking to the side with a lot of people behind her. H: A boy sleeps on the couch.	●	P: A man with a black shirt holds a baby while a blue shirted individual faces them. H: A man sits all alone on a park bench, arms folded.	●	P: Two women dressed for cold weather in jackets and gloves look at something on a cellphone screen. H: The woman is alone in her bathroom.	●
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Figure 13: Comparison between incremental and non-incremental compositional explanations for Unit 0 in the BiLSTM model.

Unit 15 – Incremental Beam Search IoU: 0.246 Explanation $(((\text{hyp:tok:sitting} \wedge \text{neighbor:pre:tok:around}) \wedge \neg \text{pre:tok:sitting}) \wedge \neg \text{pre:tag:rb}) \wedge \neg \text{neighbor:pre:tok:face})$ Samples selected by this explanation <table> <tr> <th>Premise / Hypothesis</th> <th>Ours Applies</th> </tr> <tr> <td>P: A female dancer does a center leap across a stage. H: The woman is sitting in the audience.</td> <td>●</td> </tr> <tr> <td>P: A little girl in pink boots runs down the street. H: A little girl in pink boots is sitting in a chair.</td> <td>●</td> </tr> <tr> <td>P: A blond woman is holding 2 golf balls while reaching down into a golf hole. H: The blond woman is sitting down.</td> <td>●</td> </tr> <tr> <td>P: A group of people walking arm and arm across the green grass. H: The people are sitting on the ground.</td> <td>●</td> </tr> <tr> <td>P: A man jumps over another man doing a back flip. H: Two men are sitting down, watching the game.</td> <td>●</td> </tr> </table>	Premise / Hypothesis	Ours Applies	P: A female dancer does a center leap across a stage. H: The woman is sitting in the audience.	●	P: A little girl in pink boots runs down the street. H: A little girl in pink boots is sitting in a chair.	●	P: A blond woman is holding 2 golf balls while reaching down into a golf hole. H: The blond woman is sitting down.	●	P: A group of people walking arm and arm across the green grass. H: The people are sitting on the ground.	●	P: A man jumps over another man doing a back flip. H: Two men are sitting down, watching the game.	●	Unit 15 – Our Non-Incremental Search IoU: 0.275 Explanation $((\text{hyp:tok:sitting} \wedge ((\text{pre:tok:walking} \vee \text{neighbor:pre:tok:between}) \vee \text{hyp:tok:bench})) \wedge \neg \text{oth:overlap:overlap50})$ Samples selected by this explanation <table> <tr> <th>Premise / Hypothesis</th> <th>Inc. Applies</th> </tr> <tr> <td>P: A man is a safety suit walking outside while another man in a dark suit walks into a building. H: Men are sitting in the park.</td> <td>●</td> </tr> <tr> <td>P: A female dancer does a center leap across a stage. H: The woman is sitting in the audience.</td> <td>●</td> </tr> <tr> <td>P: A group of people walking arm and arm across the green grass. H: The people are sitting on the ground.</td> <td>●</td> </tr> <tr> <td>P: A man jumps over another man doing a back flip. H: Two men are sitting down, watching the game.</td> <td>●</td> </tr> <tr> <td>P: Three people walking down a hallway at the train station. H: A few people are sitting in a dining car of a train drinking coffee.</td> <td>●</td> </tr> </table>	Premise / Hypothesis	Inc. Applies	P: A man is a safety suit walking outside while another man in a dark suit walks into a building. H: Men are sitting in the park.	●	P: A female dancer does a center leap across a stage. H: The woman is sitting in the audience.	●	P: A group of people walking arm and arm across the green grass. H: The people are sitting on the ground.	●	P: A man jumps over another man doing a back flip. H: Two men are sitting down, watching the game.	●	P: Three people walking down a hallway at the train station. H: A few people are sitting in a dining car of a train drinking coffee.	●
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Figure 14: Comparison between incremental and non-incremental compositional explanations for Unit 15 in the BiLSTM Model.

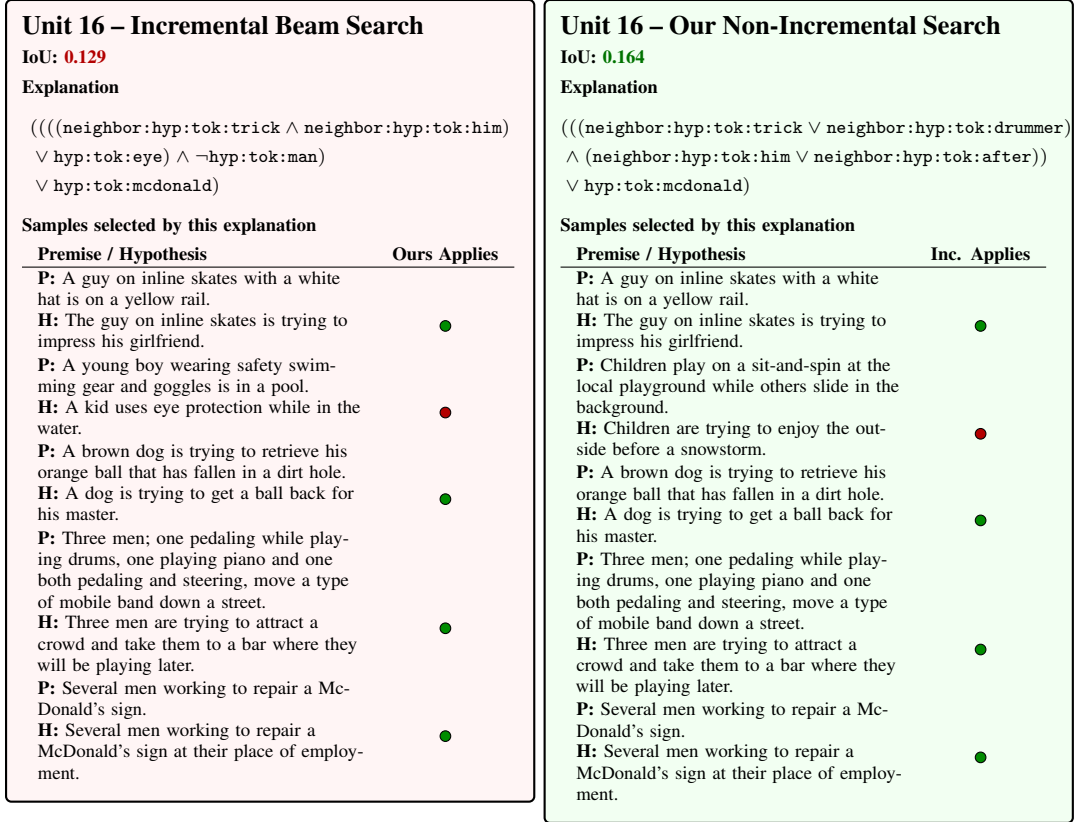


Figure 15: Comparison between incremental and non-incremental compositional explanations for Unit 16 in the BiLSTM model.

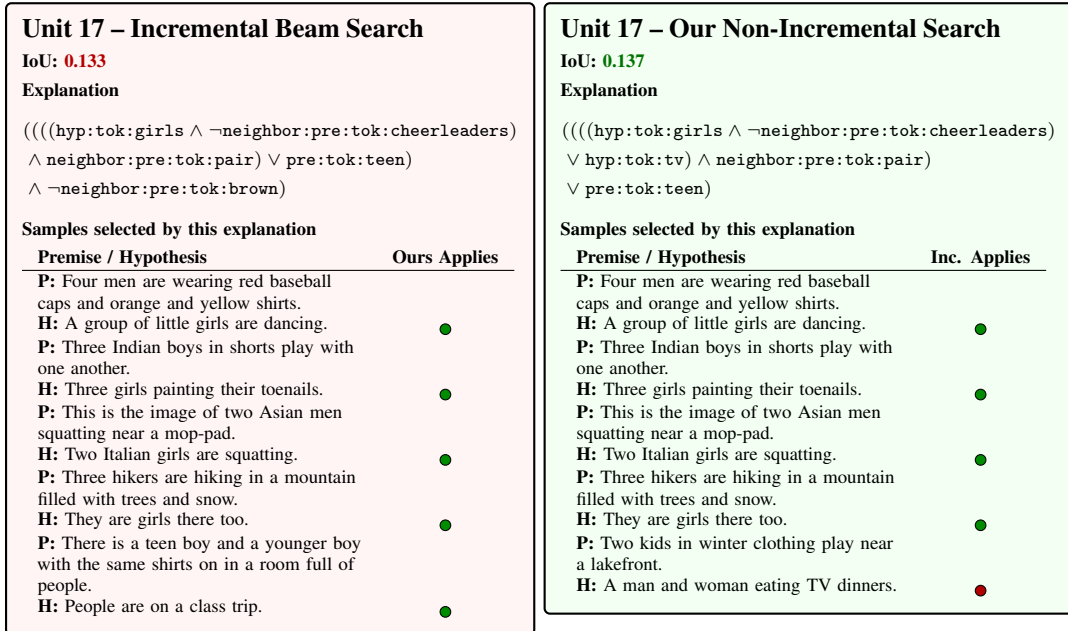


Figure 16: Comparison between incremental and non-incremental compositional explanations for Unit 17 in the BiLSTM model.

Unit 19 – Incremental Beam Search IoU: 0.109 Explanation <pre>(((hyp:tok:boys ∧ ¬pre:tok:boys) ∨ pre:tok:blows) ∧ oth:overlap:overlap25) ∨ hyp:tok:dryer)</pre> Samples selected by this explanation <table> <tr> <th>Premise / Hypothesis</th> <th>Ours Applies</th> </tr> <tr> <td>P: A blond woman talking on her cell-phone as her hair blows.</td> <td></td> </tr> <tr> <td>H: A woman talking on her cellphone.</td> <td>●</td> </tr> <tr> <td>P: Two young children in black clothing and padding perform a martial arts match where they compete against the other.</td> <td></td> </tr> <tr> <td>H: Two young boys compete in martial arts.</td> <td>●</td> </tr> <tr> <td>P: Two girls are in line about to order at a Mexican eatery.</td> <td></td> </tr> <tr> <td>H: Two boys are in line about to order at a Mexican eatery.</td> <td>●</td> </tr> <tr> <td>P: Four girls look at objects under a microscope.</td> <td></td> </tr> <tr> <td>H: Four boys look at objects under a microscope.</td> <td>●</td> </tr> <tr> <td>P: A blond woman talking on her cell-phone as her hair blows.</td> <td></td> </tr> <tr> <td>H: A woman talking on her cellphone to her husband.</td> <td>●</td> </tr> </table>	Premise / Hypothesis	Ours Applies	P: A blond woman talking on her cell-phone as her hair blows.		H: A woman talking on her cellphone.	●	P: Two young children in black clothing and padding perform a martial arts match where they compete against the other.		H: Two young boys compete in martial arts.	●	P: Two girls are in line about to order at a Mexican eatery.		H: Two boys are in line about to order at a Mexican eatery.	●	P: Four girls look at objects under a microscope.		H: Four boys look at objects under a microscope.	●	P: A blond woman talking on her cell-phone as her hair blows.		H: A woman talking on her cellphone to her husband.	●	Unit 19 – Our Non-Incremental Search IoU: 0.118 Explanation <pre>(((pre:tok:blows ∨ neighbor:hyp:tok:atv) ∨ neighbor:hyp:tok:inflatable) ∨ neighbor:hyp:tok:cheer) ∧ neighbor:hyp:tok:male)</pre> Samples selected by this explanation <table> <tr> <th>Premise / Hypothesis</th> <th>Inc. Applies</th> </tr> <tr> <td>P: Two women stop between a red smart car and a group of photographers to examine their items in hand.</td> <td></td> </tr> <tr> <td>H: Everyone pictured is male.</td> <td>●</td> </tr> <tr> <td>P: A blond woman talking on her cell-phone as her hair blows.</td> <td></td> </tr> <tr> <td>H: A woman talking on her cellphone.</td> <td>●</td> </tr> <tr> <td>P: A girls sports team putting their hands in a circle to do a bonding cheer.</td> <td></td> </tr> <tr> <td>H: Male cheerleaders talking.</td> <td>●</td> </tr> <tr> <td>P: A child is playing with a green, white and red spinning toy that turning on the ground.</td> <td></td> </tr> <tr> <td>H: A man is playing with a green, white and red spinning toy that turning on the ground.</td> <td>●</td> </tr> <tr> <td>P: Someone in a red and white uniform is driving a red and white motorcycle down a track with the front wheel in the air.</td> <td></td> </tr> <tr> <td>H: A man rides an ATV.</td> <td>●</td> </tr> </table>	Premise / Hypothesis	Inc. Applies	P: Two women stop between a red smart car and a group of photographers to examine their items in hand.		H: Everyone pictured is male.	●	P: A blond woman talking on her cell-phone as her hair blows.		H: A woman talking on her cellphone.	●	P: A girls sports team putting their hands in a circle to do a bonding cheer.		H: Male cheerleaders talking.	●	P: A child is playing with a green, white and red spinning toy that turning on the ground.		H: A man is playing with a green, white and red spinning toy that turning on the ground.	●	P: Someone in a red and white uniform is driving a red and white motorcycle down a track with the front wheel in the air.		H: A man rides an ATV.	●
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Figure 17: Comparison between incremental and non-incremental compositional explanations for Unit 19 in the BiLSTM Model.

Unit 22 – Incremental Beam Search IoU: 0.156 Explanation <pre>(((pre:tok:dress ∧ ¬neighbor:hyp:tok:female) ∧ neighbor:hyp:tok:men) ∨ pre:tok:routine) ∧ ¬pre:tag:vbn)</pre> Samples selected by this explanation <table> <tr> <th>Premise / Hypothesis</th> <th>Ours Applies</th> </tr> <tr> <td>P: Woman helps bride into dress.</td> <td></td> </tr> <tr> <td>H: A man brushes his hair.</td> <td>●</td> </tr> <tr> <td>P: A girl in a pumpkin dress sitting at a table.</td> <td></td> </tr> <tr> <td>H: There is a boy in a pumpkin dress sitting at a table.</td> <td>●</td> </tr> <tr> <td>P: A little girl in a red flowered dress with a tennis racket.</td> <td></td> </tr> <tr> <td>H: A man in red.</td> <td>●</td> </tr> <tr> <td>P: A woman in a blue dress is sweeping the floor in a cobblestone road.</td> <td></td> </tr> <tr> <td>H: A man mops the road.</td> <td>●</td> </tr> <tr> <td>P: Group of women in red swimsuits perform a dance routine in the sand at 2008 Beijing Olympics.</td> <td></td> </tr> <tr> <td>H: The men enjoy dancing in the sand.</td> <td>●</td> </tr> </table>	Premise / Hypothesis	Ours Applies	P: Woman helps bride into dress.		H: A man brushes his hair.	●	P: A girl in a pumpkin dress sitting at a table.		H: There is a boy in a pumpkin dress sitting at a table.	●	P: A little girl in a red flowered dress with a tennis racket.		H: A man in red.	●	P: A woman in a blue dress is sweeping the floor in a cobblestone road.		H: A man mops the road.	●	P: Group of women in red swimsuits perform a dance routine in the sand at 2008 Beijing Olympics.		H: The men enjoy dancing in the sand.	●	Unit 22 – Our Non-Incremental Search IoU: 0.190 Explanation <pre>(((pre:tok:dress ∨ neighbor:pre:tok:nun) ∧ (hyp:tok:man ∨ hyp:tok:boy)) ∧ ¬neighbor:hyp:tok:female)</pre> Samples selected by this explanation <table> <tr> <th>Premise / Hypothesis</th> <th>Inc. Applies</th> </tr> <tr> <td>P: A little girl and lady wearing face paint and costumes are among others holding orange balloons and an event.</td> <td></td> </tr> <tr> <td>H: A man and his son put camouflage greasepaint on their faces before they go hunting.</td> <td>●</td> </tr> <tr> <td>P: A young girl washing a lime-green car.</td> <td></td> </tr> <tr> <td>H: An old man washing a lime-green car.</td> <td>●</td> </tr> <tr> <td>P: Woman helps bride into dress.</td> <td></td> </tr> <tr> <td>H: A man brushes his hair.</td> <td>●</td> </tr> <tr> <td>P: A girl in a pumpkin dress sitting at a table.</td> <td></td> </tr> <tr> <td>H: There is a boy in a pumpkin dress sitting at a table.</td> <td>●</td> </tr> <tr> <td>P: A girl with a purple sweater and a green headband is doing classwork in a classroom.</td> <td></td> </tr> <tr> <td>H: A boy is scratching his name into his desk.</td> <td>●</td> </tr> </table>	Premise / Hypothesis	Inc. Applies	P: A little girl and lady wearing face paint and costumes are among others holding orange balloons and an event.		H: A man and his son put camouflage greasepaint on their faces before they go hunting.	●	P: A young girl washing a lime-green car.		H: An old man washing a lime-green car.	●	P: Woman helps bride into dress.		H: A man brushes his hair.	●	P: A girl in a pumpkin dress sitting at a table.		H: There is a boy in a pumpkin dress sitting at a table.	●	P: A girl with a purple sweater and a green headband is doing classwork in a classroom.		H: A boy is scratching his name into his desk.	●
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Figure 18: Comparison between incremental and non-incremental compositional explanations for Unit 22 in the BiLSTM model.

Unit 0 – Incremental Beam Search IoU: 0.136 Explanation $(((\text{hyp:tok:dancing} \wedge \text{neighbor:pre:tok:white}) \vee \text{hyp:tok:everyone}) \vee \text{hyp:tok:naked}) \wedge \neg \text{neighbor:pre:tok:hold})$ Samples selected by this explanation <table> <tr> <th>Premise / Hypothesis</th> <th>Ours Applies</th> </tr> <tr> <td>P: Four men are wearing red baseball caps and orange and yellow shirts.</td> <td></td> </tr> <tr> <td>H: A group of little girls are dancing.</td> <td>●</td> </tr> <tr> <td>P: A well built black man stands in the subway, listening to headphones.</td> <td></td> </tr> <tr> <td>H: A man is dancing for tips on the street.</td> <td>●</td> </tr> <tr> <td>P: A man in a blue windbreaker jacket, laying face down on a park bench, with a liquor bottle nestled by one of the bench's legs.</td> <td></td> </tr> <tr> <td>H: A man is dancing at the club popping bottles.</td> <td>●</td> </tr> <tr> <td>P: A man and woman and two small children are sitting in the floor with wrapped Christmas presents.</td> <td></td> </tr> <tr> <td>H: Everyone in the photo are related.</td> <td>●</td> </tr> <tr> <td>P: A man and woman hug.</td> <td></td> </tr> <tr> <td>H: Everyone is standing apart.</td> <td>●</td> </tr> </table>	Premise / Hypothesis	Ours Applies	P: Four men are wearing red baseball caps and orange and yellow shirts.		H: A group of little girls are dancing.	●	P: A well built black man stands in the subway, listening to headphones.		H: A man is dancing for tips on the street.	●	P: A man in a blue windbreaker jacket, laying face down on a park bench, with a liquor bottle nestled by one of the bench's legs.		H: A man is dancing at the club popping bottles.	●	P: A man and woman and two small children are sitting in the floor with wrapped Christmas presents.		H: Everyone in the photo are related.	●	P: A man and woman hug.		H: Everyone is standing apart.	●	Unit 0 – Our Non-Incremental Search IoU: 0.141 Explanation $(((\text{hyp:tok:dancing} \vee \text{pre:tok:desk}) \wedge \text{neighbor:pre:tok:white}) \vee \text{hyp:tok:everyone}) \vee \text{pre:tok:antique})$ Samples selected by this explanation <table> <tr> <th>Premise / Hypothesis</th> <th>Inc. Applies</th> </tr> <tr> <td>P: Four men are wearing red baseball caps and orange and yellow shirts.</td> <td></td> </tr> <tr> <td>H: A group of little girls are dancing.</td> <td>●</td> </tr> <tr> <td>P: A well built black man stands in the subway, listening to headphones.</td> <td></td> </tr> <tr> <td>H: A man is dancing for tips on the street.</td> <td>●</td> </tr> <tr> <td>P: A man in a blue windbreaker jacket, laying face down on a park bench, with a liquor bottle nestled by one of the bench's legs.</td> <td></td> </tr> <tr> <td>H: A man is dancing at the club popping bottles.</td> <td>●</td> </tr> <tr> <td>P: A man and woman and two small children are sitting in the floor with wrapped Christmas presents.</td> <td></td> </tr> <tr> <td>H: Everyone in the photo are related.</td> <td>●</td> </tr> <tr> <td>P: A girl wearing a red shirt and jeans is sitting in a rocking chair at a cluttered desk, while another girl wearing a black t-shirt and jogging pants is sitting on the floor writing on a piece of paper with other papers arranged around her.</td> <td></td> </tr> <tr> <td>H: Two girls flying a kite in the park.</td> <td>●</td> </tr> </table>	Premise / Hypothesis	Inc. Applies	P: Four men are wearing red baseball caps and orange and yellow shirts.		H: A group of little girls are dancing.	●	P: A well built black man stands in the subway, listening to headphones.		H: A man is dancing for tips on the street.	●	P: A man in a blue windbreaker jacket, laying face down on a park bench, with a liquor bottle nestled by one of the bench's legs.		H: A man is dancing at the club popping bottles.	●	P: A man and woman and two small children are sitting in the floor with wrapped Christmas presents.		H: Everyone in the photo are related.	●	P: A girl wearing a red shirt and jeans is sitting in a rocking chair at a cluttered desk, while another girl wearing a black t-shirt and jogging pants is sitting on the floor writing on a piece of paper with other papers arranged around her.		H: Two girls flying a kite in the park.	●
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Figure 19: Comparison between incremental and non-incremental compositional explanations for Unit 0 in the BERT model.

Unit 2 – Incremental Beam Search IoU: 0.141 Explanation (((((pre:tok:gloves $\vee$ pre:tok:backs) $\wedge$ neighbor:pre:tok:shirt) $\vee$ pre:tok:!) $\vee$ pre:tok:denim) Samples selected by this explanation <table> <tr> <th>Premise / Hypothesis</th> <th>Ours Applies</th> </tr> <tr> <td>P: Man with dark shirt in work gloves and a helmet that has a yellow flashlight attached.</td> <td></td> </tr> <tr> <td>H: A woman handing out water.</td> <td>●</td> </tr> <tr> <td>P: Making something tasty hope you like lemons!</td> <td></td> </tr> <tr> <td>H: A lady is sewing up a hole in a dress.</td> <td>●</td> </tr> <tr> <td>P: A crowd of people with a lady with a bright orange shirt and another lady in black have their backs to the camera.</td> <td></td> </tr> <tr> <td>H: Three girls try to buy candy in a grocery store.</td> <td>●</td> </tr> <tr> <td>P: Man with dark shirt in work gloves and a helmet that has a yellow flashlight attached.</td> <td></td> </tr> <tr> <td>H: An rescue worker looking for injured people.</td> <td>●</td> </tr> <tr> <td>P: Making something tasty hope you like lemons!</td> <td></td> </tr> <tr> <td>H: A woman is making a lemon meringue pie.</td> <td>●</td> </tr> </table>	Premise / Hypothesis	Ours Applies	P: Man with dark shirt in work gloves and a helmet that has a yellow flashlight attached.		H: A woman handing out water.	●	P: Making something tasty hope you like lemons!		H: A lady is sewing up a hole in a dress.	●	P: A crowd of people with a lady with a bright orange shirt and another lady in black have their backs to the camera.		H: Three girls try to buy candy in a grocery store.	●	P: Man with dark shirt in work gloves and a helmet that has a yellow flashlight attached.		H: An rescue worker looking for injured people.	●	P: Making something tasty hope you like lemons!		H: A woman is making a lemon meringue pie.	●	Unit 2 – Our Non-Incremental Search IoU: 0.140 Explanation (((((pre:tok:you $\vee$ pre:tok:denim) $\wedge$ $\neg$pre:tag:vbz) $\vee$ pre:tok:backs) $\vee$ hyp:tok:stay) Samples selected by this explanation <table> <tr> <th>Premise / Hypothesis</th> <th>Inc. Applies</th> </tr> <tr> <td>P: Making something tasty hope you like lemons!</td> <td></td> </tr> <tr> <td>H: A lady is sewing up a hole in a dress.</td> <td>●</td> </tr> <tr> <td>P: A crowd of people with a lady with a bright orange shirt and another lady in black have their backs to the camera.</td> <td></td> </tr> <tr> <td>H: Three girls try to buy candy in a grocery store.</td> <td>●</td> </tr> <tr> <td>P: Making something tasty hope you like lemons!</td> <td></td> </tr> <tr> <td>H: A woman is making a lemon meringue pie.</td> <td>●</td> </tr> <tr> <td>P: A small dog wearing a denim miniskirt.</td> <td></td> </tr> <tr> <td>H: A dog is wearing a skirt.</td> <td>●</td> </tr> <tr> <td>P: A woman walking, smiling and holding a shopping bag.</td> <td></td> </tr> <tr> <td>H: Some women go shopping, while men stay at home.</td> <td>●</td> </tr> </table>	Premise / Hypothesis	Inc. Applies	P: Making something tasty hope you like lemons!		H: A lady is sewing up a hole in a dress.	●	P: A crowd of people with a lady with a bright orange shirt and another lady in black have their backs to the camera.		H: Three girls try to buy candy in a grocery store.	●	P: Making something tasty hope you like lemons!		H: A woman is making a lemon meringue pie.	●	P: A small dog wearing a denim miniskirt.		H: A dog is wearing a skirt.	●	P: A woman walking, smiling and holding a shopping bag.		H: Some women go shopping, while men stay at home.	●
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Figure 20: Comparison between incremental and non-incremental compositional explanations for Unit 2 in the BERT model.

Unit 3 – Incremental Beam Search IoU: 0.118 Explanation (((((pre:tok:smile $\vee$ hyp:tok:cheerleaders) $\vee$ pre:tok:pretty) $\vee$ neighbor:hyp:tok:upset) $\wedge$ $\neg$hyp:tok:on) Samples selected by this explanation <table> <tr> <th>Premise / Hypothesis</th> <th>Ours Applies</th> </tr> <tr> <td>P: Young brunette woman smiling in front of bushes.</td> <td></td> </tr> <tr> <td>H: The man looks very upset.</td> <td>●</td> </tr> <tr> <td>P: A woman is smiling at a baby in a stroller in a meadow.</td> <td></td> </tr> <tr> <td>H: A woman is angry at the baby.</td> <td>●</td> </tr> <tr> <td>P: A young girl in a pretty pink and white blouse and shorts is participating in a soccer game with other children.</td> <td></td> </tr> <tr> <td>H: A man is making a candy sculpture out of peppermints.</td> <td>●</td> </tr> <tr> <td>P: A girls sports team putting their hands in a circle to do a bonding cheer.</td> <td></td> </tr> <tr> <td>H: Male cheerleaders talking.</td> <td>●</td> </tr> <tr> <td>P: Two girls smile for the camera.</td> <td></td> </tr> <tr> <td>H: Three men are smoking cigars.</td> <td>●</td> </tr> </table>	Premise / Hypothesis	Ours Applies	P: Young brunette woman smiling in front of bushes.		H: The man looks very upset.	●	P: A woman is smiling at a baby in a stroller in a meadow.		H: A woman is angry at the baby.	●	P: A young girl in a pretty pink and white blouse and shorts is participating in a soccer game with other children.		H: A man is making a candy sculpture out of peppermints.	●	P: A girls sports team putting their hands in a circle to do a bonding cheer.		H: Male cheerleaders talking.	●	P: Two girls smile for the camera.		H: Three men are smoking cigars.	●	Unit 3 – Our Non-Incremental Search IoU: 0.120 Explanation (((((pre:tok:smile $\wedge$ $\neg$hyp:tag:nn) $\vee$ hyp:tok:cheerleaders) $\vee$ pre:tok:pretty) $\vee$ neighbor:hyp:tok:upset) Samples selected by this explanation <table> <tr> <th>Premise / Hypothesis</th> <th>Inc. Applies</th> </tr> <tr> <td>P: Young brunette woman smiling in front of bushes.</td> <td></td> </tr> <tr> <td>H: The man looks very upset.</td> <td>●</td> </tr> <tr> <td>P: A woman is smiling at a baby in a stroller in a meadow.</td> <td></td> </tr> <tr> <td>H: A woman is angry at the baby.</td> <td>●</td> </tr> <tr> <td>P: A young girl in a pretty pink and white blouse and shorts is participating in a soccer game with other children.</td> <td></td> </tr> <tr> <td>H: A man is making a candy sculpture out of peppermints.</td> <td>●</td> </tr> <tr> <td>P: A girls sports team putting their hands in a circle to do a bonding cheer.</td> <td></td> </tr> <tr> <td>H: Male cheerleaders talking.</td> <td>●</td> </tr> <tr> <td>P: Two girls smile for the camera.</td> <td></td> </tr> <tr> <td>H: Three men are smoking cigars.</td> <td>●</td> </tr> </table>	Premise / Hypothesis	Inc. Applies	P: Young brunette woman smiling in front of bushes.		H: The man looks very upset.	●	P: A woman is smiling at a baby in a stroller in a meadow.		H: A woman is angry at the baby.	●	P: A young girl in a pretty pink and white blouse and shorts is participating in a soccer game with other children.		H: A man is making a candy sculpture out of peppermints.	●	P: A girls sports team putting their hands in a circle to do a bonding cheer.		H: Male cheerleaders talking.	●	P: Two girls smile for the camera.		H: Three men are smoking cigars.	●
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Figure 21: Comparison between incremental and non-incremental compositional explanations for Unit 3 in the BERT model.



Figure 22: Comparison between incremental and non-incremental compositional explanations for Unit 4 in the BERT model.

Unit 5 – Incremental Beam Search IoU: 0.132 Explanation $(((\text{hyp:tok:swimming} \wedge \text{hyp:tok:ocean}) \vee \text{pre:tok:crafts}) \vee \text{hyp:tok:pictured}) \vee \text{hyp:tok:swims})$ Samples selected by this explanation <table> <tr> <th>Premise / Hypothesis</th> <th>Ours Applies</th> </tr> <tr> <td>P: Two women stop between a red smart car and a group of photographers to examine their items in hand.</td> <td></td> </tr> <tr> <td>H: Everyone pictured is male.</td> <td>●</td> </tr> <tr> <td>P: A woman in a jacket and scarf looking a table that a man in a black shirt is cooking something on.</td> <td></td> </tr> <tr> <td>H: A man and woman are swimming in the ocean.</td> <td>●</td> </tr> <tr> <td>P: People shopping at the marketplace.</td> <td></td> </tr> <tr> <td>H: A woman is swimming in the ocean.</td> <td>●</td> </tr> <tr> <td>P: A young boy reaches for and touches the propeller of a vintage aircraft.</td> <td></td> </tr> <tr> <td>H: A young boy swims in his pool.</td> <td>●</td> </tr> <tr> <td>P: Three women and one man are sitting on a ledge facing away towards a large window.</td> <td></td> </tr> <tr> <td>H: People are swimming in the ocean.</td> <td>●</td> </tr> </table>	Premise / Hypothesis	Ours Applies	P: Two women stop between a red smart car and a group of photographers to examine their items in hand.		H: Everyone pictured is male.	●	P: A woman in a jacket and scarf looking a table that a man in a black shirt is cooking something on.		H: A man and woman are swimming in the ocean.	●	P: People shopping at the marketplace.		H: A woman is swimming in the ocean.	●	P: A young boy reaches for and touches the propeller of a vintage aircraft.		H: A young boy swims in his pool.	●	P: Three women and one man are sitting on a ledge facing away towards a large window.		H: People are swimming in the ocean.	●	Unit 5 – Our Non-Incremental Search IoU: 0.185 Explanation $((\text{hyp:tok:swimming} \vee \text{hyp:tok:play}) \wedge ((\text{neighbor:hyp:tok:shore} \vee \text{neighbor:hyp:tok:alone}) \vee \text{pre:tok:walking}))$ Samples selected by this explanation <table> <tr> <th>Premise / Hypothesis</th> <th>Inc. Applies</th> </tr> <tr> <td>P: A woman in a jacket and scarf looking a table that a man in a black shirt is cooking something on.</td> <td></td> </tr> <tr> <td>H: A man and woman are swimming in the ocean.</td> <td>●</td> </tr> <tr> <td>P: Lady vendor selling her goods, one man standing on a post, and two men in a wholesale store.</td> <td></td> </tr> <tr> <td>H: A dog sits alone next to a swimming pool.</td> <td>●</td> </tr> <tr> <td>P: People walking through dirt.</td> <td></td> </tr> <tr> <td>H: People are swimming.</td> <td>●</td> </tr> <tr> <td>P: People shopping at the marketplace.</td> <td></td> </tr> <tr> <td>H: A woman is swimming in the ocean.</td> <td>●</td> </tr> <tr> <td>P: Three women and one man are sitting on a ledge facing away towards a large window.</td> <td></td> </tr> <tr> <td>H: People are swimming in the ocean.</td> <td>●</td> </tr> </table>	Premise / Hypothesis	Inc. Applies	P: A woman in a jacket and scarf looking a table that a man in a black shirt is cooking something on.		H: A man and woman are swimming in the ocean.	●	P: Lady vendor selling her goods, one man standing on a post, and two men in a wholesale store.		H: A dog sits alone next to a swimming pool.	●	P: People walking through dirt.		H: People are swimming.	●	P: People shopping at the marketplace.		H: A woman is swimming in the ocean.	●	P: Three women and one man are sitting on a ledge facing away towards a large window.		H: People are swimming in the ocean.	●
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Figure 23: Comparison between incremental and non-incremental compositional explanations for Unit 5 in the BERT model.

Unit 1 – Incremental Beam Search IoU: 0.179 Explanation $(((\text{hyp:tok:moon} \vee \text{hyp:tok:empty}) \vee \text{pre:tok:types}) \wedge \neg \text{pre:tag:cd}) \vee \text{hyp:tok:nothing})$ Samples selected by this explanation <table> <tr> <th>Premise / Hypothesis</th> <th>Ours Applies</th> </tr> <tr> <td>P: Young blond girl holding chopsticks looks with red eyes at camera while another girl in the background looks at food.</td> <td></td> </tr> <tr> <td>H: The girl is alone in the middle of the ocean on top of the moon.</td> <td>●</td> </tr> <tr> <td>P: There is a room full of pictures all in the wall and a woman in a coat is looking back over her shoulders strangely.</td> <td></td> </tr> <tr> <td>H: The room is empty.</td> <td>●</td> </tr> <tr> <td>P: A young man in a black shirt and an elderly woman in a pink shirt sitting on a train together.</td> <td></td> </tr> <tr> <td>H: The man is asleep in the bathtub on the moon.</td> <td>●</td> </tr> <tr> <td>P: A lady in a business suit sits at a desk and types on her keyboard.</td> <td></td> </tr> <tr> <td>H: The boy liked cowboys.</td> <td>●</td> </tr> <tr> <td>P: Interior scene of fairly full tour bus.</td> <td></td> </tr> <tr> <td>H: The bus is empty.</td> <td>●</td> </tr> </table>	Premise / Hypothesis	Ours Applies	P: Young blond girl holding chopsticks looks with red eyes at camera while another girl in the background looks at food.		H: The girl is alone in the middle of the ocean on top of the moon.	●	P: There is a room full of pictures all in the wall and a woman in a coat is looking back over her shoulders strangely.		H: The room is empty.	●	P: A young man in a black shirt and an elderly woman in a pink shirt sitting on a train together.		H: The man is asleep in the bathtub on the moon.	●	P: A lady in a business suit sits at a desk and types on her keyboard.		H: The boy liked cowboys.	●	P: Interior scene of fairly full tour bus.		H: The bus is empty.	●	Unit 1 – Our Non-Incremental Search IoU: 0.185 Explanation $(((\text{hyp:tok:moon} \vee \text{hyp:tok:nothing}) \vee \text{hyp:tok:empty} \wedge \neg \text{hyp:tag:in})) \vee \text{hyp:tok:straight})$ Samples selected by this explanation <table> <tr> <th>Premise / Hypothesis</th> <th>Inc. Applies</th> </tr> <tr> <td>P: Young blond girl holding chopsticks looks with red eyes at camera while another girl in the background looks at food.</td> <td></td> </tr> <tr> <td>H: The girl is alone in the middle of the ocean on top of the moon.</td> <td>●</td> </tr> <tr> <td>P: There is a room full of pictures all in the wall and a woman in a coat is looking back over her shoulders strangely.</td> <td></td> </tr> <tr> <td>H: The room is empty.</td> <td>●</td> </tr> <tr> <td>P: A young man in a black shirt and an elderly woman in a pink shirt sitting on a train together.</td> <td></td> </tr> <tr> <td>H: The man is asleep in the bathtub on the moon.</td> <td>●</td> </tr> <tr> <td>P: Interior scene of fairly full tour bus.</td> <td></td> </tr> <tr> <td>H: The bus is empty.</td> <td>●</td> </tr> <tr> <td>P: A woman in glittery face-paint and a sequined cap parades with several similarly-dressed individuals in shades of orange and brown.</td> <td></td> </tr> <tr> <td>H: The woman is swimming alone in her indoor pool on the moon.</td> <td>●</td> </tr> </table>	Premise / Hypothesis	Inc. Applies	P: Young blond girl holding chopsticks looks with red eyes at camera while another girl in the background looks at food.		H: The girl is alone in the middle of the ocean on top of the moon.	●	P: There is a room full of pictures all in the wall and a woman in a coat is looking back over her shoulders strangely.		H: The room is empty.	●	P: A young man in a black shirt and an elderly woman in a pink shirt sitting on a train together.		H: The man is asleep in the bathtub on the moon.	●	P: Interior scene of fairly full tour bus.		H: The bus is empty.	●	P: A woman in glittery face-paint and a sequined cap parades with several similarly-dressed individuals in shades of orange and brown.		H: The woman is swimming alone in her indoor pool on the moon.	●
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Figure 24: Comparison between incremental and non-incremental compositional explanations for Unit 1 in the LLaMA model.

Unit 2 – Incremental Beam Search IoU: 0.148 Explanation <pre>(((((hyp:tok:naked ∨ hyp:tok:sun) ∧ ¬neighbor:pre:tok:man) ∨ pre:tok:airport) ∧ ¬neighbor:pre:tok:at))</pre> Samples selected by this explanation <table> <tr> <th>Premise / Hypothesis</th> <th>Ours Applies</th> </tr> <tr> <td>P: The tops of a large number of balloons are visible behind a wall, with a large high rise building in the background.</td> <td></td> </tr> <tr> <td>H: A man and woman watch the sun set over the ocean.</td> <td>●</td> </tr> <tr> <td>P: A person wearing a suit is walking through an airport.</td> <td></td> </tr> <tr> <td>H: A woman sits in the lobby waiting on the doctor.</td> <td>●</td> </tr> <tr> <td>P: People dressed in colorful outfits are gathered together.</td> <td></td> </tr> <tr> <td>H: The man and women walked naked along the secluded beach.</td> <td>●</td> </tr> <tr> <td>P: People dressed in costumes walking down a street.</td> <td></td> </tr> <tr> <td>H: A naked man walking down the street.</td> <td>●</td> </tr> <tr> <td>P: Child inside of large snow fort.</td> <td></td> </tr> <tr> <td>H: A boy sweating in the sun.</td> <td>●</td> </tr> </table>	Premise / Hypothesis	Ours Applies	P: The tops of a large number of balloons are visible behind a wall, with a large high rise building in the background.		H: A man and woman watch the sun set over the ocean.	●	P: A person wearing a suit is walking through an airport.		H: A woman sits in the lobby waiting on the doctor.	●	P: People dressed in colorful outfits are gathered together.		H: The man and women walked naked along the secluded beach.	●	P: People dressed in costumes walking down a street.		H: A naked man walking down the street.	●	P: Child inside of large snow fort.		H: A boy sweating in the sun.	●	Unit 2 – Our Non-Incremental Search IoU: 0.164 Explanation <pre>(((((neighbor:pre:tok:snowboarder ∨ pre:tok:person) ∧ neighbor:hyp:tok:man) ∧ ¬pre:tag:cc) ∧ ¬neighbor:pre:tok:men))</pre> Samples selected by this explanation <table> <tr> <th>Premise / Hypothesis</th> <th>Inc. Applies</th> </tr> <tr> <td>P: A person performing a snowboarding trick on a high mountain.</td> <td></td> </tr> <tr> <td>H: A young man is picking up his girlfriend for a date.</td> <td>●</td> </tr> <tr> <td>P: A person is climbing a cliff wall, over a rocky shore, using a rope.</td> <td></td> </tr> <tr> <td>H: A man is swimming to his favorite beach.</td> <td>●</td> </tr> <tr> <td>P: A person in gray snowboarding down a hill.</td> <td></td> </tr> <tr> <td>H: The snowboarder is a man.</td> <td>●</td> </tr> <tr> <td>P: A person wearing a suit is walking through an airport.</td> <td></td> </tr> <tr> <td>H: A woman sits in the lobby waiting on the doctor.</td> <td>●</td> </tr> <tr> <td>P: A person is riding a colorful dirt bike on dirt hills.</td> <td></td> </tr> <tr> <td>H: A boy plays with his remote control car in his living room.</td> <td>●</td> </tr> </table>	Premise / Hypothesis	Inc. Applies	P: A person performing a snowboarding trick on a high mountain.		H: A young man is picking up his girlfriend for a date.	●	P: A person is climbing a cliff wall, over a rocky shore, using a rope.		H: A man is swimming to his favorite beach.	●	P: A person in gray snowboarding down a hill.		H: The snowboarder is a man.	●	P: A person wearing a suit is walking through an airport.		H: A woman sits in the lobby waiting on the doctor.	●	P: A person is riding a colorful dirt bike on dirt hills.		H: A boy plays with his remote control car in his living room.	●
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Figure 25: Comparison between incremental and non-incremental compositional explanations for Unit 2 in the LLaMA model.

Unit 3 – Incremental Beam Search IoU: 0.224 Explanation <pre>(((((hyp:tok:picture ∧ hyp:tag:ex) ∨ hyp:tok:humans) ∨ hyp:tok:least) ∧ ¬neighbor:pre:tok:into))</pre> Samples selected by this explanation <table> <tr> <th>Premise / Hypothesis</th> <th>Ours Applies</th> </tr> <tr> <td>P: Sidewalk vendor dressed in white shirt brown pants sells grilled corn.</td> <td></td> </tr> <tr> <td>H: There is a person in this picture.</td> <td>●</td> </tr> <tr> <td>P: A group of people in costumes are holding their arms in the air in the manner of the finale of a musical number.</td> <td></td> </tr> <tr> <td>H: There are some humans wearing outfits.</td> <td>●</td> </tr> <tr> <td>P: Seagulls are flying above the crashing waves.</td> <td></td> </tr> <tr> <td>H: There are flying animals in this picture.</td> <td>●</td> </tr> <tr> <td>P: Many people have painted faces at night.</td> <td></td> </tr> <tr> <td>H: There are at least 2 people.</td> <td>●</td> </tr> <tr> <td>P: Two people sit facing away in a downtown scene with a motorcycle parked in front of a pool.</td> <td></td> </tr> <tr> <td>H: There are 2 people in this picture, as well as a two-wheeled vehicle.</td> <td>●</td> </tr> </table>	Premise / Hypothesis	Ours Applies	P: Sidewalk vendor dressed in white shirt brown pants sells grilled corn.		H: There is a person in this picture.	●	P: A group of people in costumes are holding their arms in the air in the manner of the finale of a musical number.		H: There are some humans wearing outfits.	●	P: Seagulls are flying above the crashing waves.		H: There are flying animals in this picture.	●	P: Many people have painted faces at night.		H: There are at least 2 people.	●	P: Two people sit facing away in a downtown scene with a motorcycle parked in front of a pool.		H: There are 2 people in this picture, as well as a two-wheeled vehicle.	●	Unit 3 – Our Non-Incremental Search IoU: 0.236 Explanation <pre>((hyp:tok:least ∨ (hyp:tok:humans ∧ pre:tag:jj)) ∨ (hyp:tok:picture ∧ hyp:tag:ex))</pre> Samples selected by this explanation <table> <tr> <th>Premise / Hypothesis</th> <th>Inc. Applies</th> </tr> <tr> <td>P: Sidewalk vendor dressed in white shirt brown pants sells grilled corn.</td> <td></td> </tr> <tr> <td>H: There is a person in this picture.</td> <td>●</td> </tr> <tr> <td>P: A group of people in costumes are holding their arms in the air in the manner of the finale of a musical number.</td> <td></td> </tr> <tr> <td>H: There are some humans wearing outfits.</td> <td>●</td> </tr> <tr> <td>P: Seagulls are flying above the crashing waves.</td> <td></td> </tr> <tr> <td>H: There are flying animals in this picture.</td> <td>●</td> </tr> <tr> <td>P: Many people have painted faces at night.</td> <td></td> </tr> <tr> <td>H: There are at least 2 people.</td> <td>●</td> </tr> <tr> <td>P: Two people sit facing away in a downtown scene with a motorcycle parked in front of a pool.</td> <td></td> </tr> <tr> <td>H: There are 2 people in this picture, as well as a two-wheeled vehicle.</td> <td>●</td> </tr> </table>	Premise / Hypothesis	Inc. Applies	P: Sidewalk vendor dressed in white shirt brown pants sells grilled corn.		H: There is a person in this picture.	●	P: A group of people in costumes are holding their arms in the air in the manner of the finale of a musical number.		H: There are some humans wearing outfits.	●	P: Seagulls are flying above the crashing waves.		H: There are flying animals in this picture.	●	P: Many people have painted faces at night.		H: There are at least 2 people.	●	P: Two people sit facing away in a downtown scene with a motorcycle parked in front of a pool.		H: There are 2 people in this picture, as well as a two-wheeled vehicle.	●
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Figure 26: Comparison between incremental and non-incremental compositional explanations for Unit 3 in the LLaMA model.

Unit 4 – Incremental Beam Search IoU: 0.159 Explanation <pre>((((pre:tok:running ∧ ¬neighbor:hyp:tok:run) ∧ neighbor:hyp:tok:women) ∨ pre:tok:skydiving) ∨ hyp:tok:cook)</pre> Samples selected by this explanation <table> <tr> <th>Premise / Hypothesis</th> <th>Ours Applies</th> </tr> <tr> <td>P: A group of grimacing males is running on green grass, several of them are wearing Nike sneakers.</td> <td></td> </tr> <tr> <td>H: The men are all wearing sandals.</td> <td>●</td> </tr> <tr> <td>P: A child and small dog running.</td> <td></td> </tr> <tr> <td>H: A man eats a carrot.</td> <td>●</td> </tr> <tr> <td>P: The man in the black and white shirt is running with a ball.</td> <td></td> </tr> <tr> <td>H: The man in the black and white shirt is sitting with a book.</td> <td>●</td> </tr> <tr> <td>P: Two yellow dogs are running side by side down a road.</td> <td></td> </tr> <tr> <td>H: The 2 dogs cook pizza.</td> <td>●</td> </tr> <tr> <td>P: A girl in uniform running through the water fountain gushing water from the tiles.</td> <td></td> </tr> <tr> <td>H: A girls is happy.</td> <td>●</td> </tr> </table>	Premise / Hypothesis	Ours Applies	P: A group of grimacing males is running on green grass, several of them are wearing Nike sneakers.		H: The men are all wearing sandals.	●	P: A child and small dog running.		H: A man eats a carrot.	●	P: The man in the black and white shirt is running with a ball.		H: The man in the black and white shirt is sitting with a book.	●	P: Two yellow dogs are running side by side down a road.		H: The 2 dogs cook pizza.	●	P: A girl in uniform running through the water fountain gushing water from the tiles.		H: A girls is happy.	●	Unit 4 – Our Non-Incremental Search IoU: 0.183 Explanation <pre>((((pre:tok:running ∨ hyp:tok:boots) ∧ (neighbor:hyp:tok:boys ∨ hyp:tok:sitting)) ∧ ¬neighbor:hyp:tok:runner)</pre> Samples selected by this explanation <table> <tr> <th>Premise / Hypothesis</th> <th>Inc. Applies</th> </tr> <tr> <td>P: A girl in green is running behind a boy in red in a sports arena whilst a girl in pink is running in the background.</td> <td></td> </tr> <tr> <td>H: Some kids sit on a rollercoaster.</td> <td>●</td> </tr> <tr> <td>P: A little girl in pink boots runs down the street.</td> <td></td> </tr> <tr> <td>H: A little girl in pink boots is sitting in a chair.</td> <td>●</td> </tr> <tr> <td>P: Girl with helmet riding a brown horse.</td> <td></td> </tr> <tr> <td>H: The girl is wearing tall boots.</td> <td>●</td> </tr> <tr> <td>P: The man in the black and white shirt is running with a ball.</td> <td></td> </tr> <tr> <td>H: The man in the black and white shirt is sitting with a book.</td> <td>●</td> </tr> <tr> <td>P: A girl leaps through the air while running on the riverbank.</td> <td></td> </tr> <tr> <td>H: The girl is wearing pants.</td> <td>●</td> </tr> </table>	Premise / Hypothesis	Inc. Applies	P: A girl in green is running behind a boy in red in a sports arena whilst a girl in pink is running in the background.		H: Some kids sit on a rollercoaster.	●	P: A little girl in pink boots runs down the street.		H: A little girl in pink boots is sitting in a chair.	●	P: Girl with helmet riding a brown horse.		H: The girl is wearing tall boots.	●	P: The man in the black and white shirt is running with a ball.		H: The man in the black and white shirt is sitting with a book.	●	P: A girl leaps through the air while running on the riverbank.		H: The girl is wearing pants.	●
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Figure 27: Comparison between incremental and non-incremental compositional explanations for Unit 4 in the LLaMA model.

Unit 8 – Incremental Beam Search IoU: 0.135 Explanation <pre>((((pre:tok:swimming ∨ pre:tok:wave) ∧ ¬oth:overlap:overlap50) ∧ ¬neighbor:hyp:tok:others ∧ ¬neighbor:hyp:tok:playground)</pre> Samples selected by this explanation <table> <tr> <th>Premise / Hypothesis</th> <th>Ours Applies</th> </tr> <tr> <td>P: Five young boys in swim trunks jump into a swimming pool at the same time.</td> <td></td> </tr> <tr> <td>H: A group of young boys studying at the library.</td> <td>●</td> </tr> <tr> <td>P: A person surfs through a wave as it crashes down.</td> <td></td> </tr> <tr> <td>H: A person is outdoors.</td> <td>●</td> </tr> <tr> <td>P: A lady wearing a red and white polka dot swim cap is swimming in a pool.</td> <td></td> </tr> <tr> <td>H: Woman sits on the sidelines of a basketball game.</td> <td>●</td> </tr> <tr> <td>P: A person surfs through a wave as it crashes down.</td> <td></td> </tr> <tr> <td>H: A person is lying on the beach.</td> <td>●</td> </tr> <tr> <td>P: A brown dog and a black dog in the edge of the ocean with a wave under them boats are on the water in the back-ground.</td> <td></td> </tr> <tr> <td>H: The pets are sleeping on the grass.</td> <td>●</td> </tr> </table>	Premise / Hypothesis	Ours Applies	P: Five young boys in swim trunks jump into a swimming pool at the same time.		H: A group of young boys studying at the library.	●	P: A person surfs through a wave as it crashes down.		H: A person is outdoors.	●	P: A lady wearing a red and white polka dot swim cap is swimming in a pool.		H: Woman sits on the sidelines of a basketball game.	●	P: A person surfs through a wave as it crashes down.		H: A person is lying on the beach.	●	P: A brown dog and a black dog in the edge of the ocean with a wave under them boats are on the water in the back-ground.		H: The pets are sleeping on the grass.	●	Unit 8 – Our Non-Incremental Search IoU: 0.159 Explanation <pre>((((pre:tok:swimming ∧ ¬oth:overlap:overlap25) ∧ ¬pre:tag:rb) ∨ (pre:tok:wave ∧ pre:tag:prp))</pre> Samples selected by this explanation <table> <tr> <th>Premise / Hypothesis</th> <th>Inc. Applies</th> </tr> <tr> <td>P: A person surfs through a wave as it crashes down.</td> <td></td> </tr> <tr> <td>H: A person is outdoors.</td> <td>●</td> </tr> <tr> <td>P: A lady wearing a red and white polka dot swim cap is swimming in a pool.</td> <td></td> </tr> <tr> <td>H: Woman sits on the sidelines of a basketball game.</td> <td>●</td> </tr> <tr> <td>P: A person surfs through a wave as it crashes down.</td> <td></td> </tr> <tr> <td>H: A person is lying on the beach.</td> <td>●</td> </tr> <tr> <td>P: A brown dog and a black dog in the edge of the ocean with a wave under them boats are on the water in the back-ground.</td> <td></td> </tr> <tr> <td>H: The pets are sleeping on the grass.</td> <td>●</td> </tr> <tr> <td>P: Two little girls on a hot summer day playing in a swimming pool.</td> <td></td> </tr> <tr> <td>H: The girl is alone in her bedroom asleep.</td> <td>●</td> </tr> </table>	Premise / Hypothesis	Inc. Applies	P: A person surfs through a wave as it crashes down.		H: A person is outdoors.	●	P: A lady wearing a red and white polka dot swim cap is swimming in a pool.		H: Woman sits on the sidelines of a basketball game.	●	P: A person surfs through a wave as it crashes down.		H: A person is lying on the beach.	●	P: A brown dog and a black dog in the edge of the ocean with a wave under them boats are on the water in the back-ground.		H: The pets are sleeping on the grass.	●	P: Two little girls on a hot summer day playing in a swimming pool.		H: The girl is alone in her bedroom asleep.	●
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Figure 28: Comparison between incremental and non-incremental compositional explanations for Unit 8 in the LLaMA model.

Unit 9 – Incremental Beam Search IoU: 0.342 Explanation $(((\text{hyp:tok:sleeping} \vee \text{hyp:tok:asleep}) \wedge \text{neighbor:hyp:tok:in}) \vee \text{hyp:tok:sleeps}) \wedge \neg \text{neighbor:hyp:tok:women})$ Samples selected by this explanation <table> <tr> <th>Premise / Hypothesis</th> <th>Ours Applies</th> </tr> <tr> <td>P: A woman in a white hat and red dress holding hands with a man in a black jacket with a flower in his upper pocket.</td> <td></td> </tr> <tr> <td>H: The woman is alone and asleep in the car on the moon.</td> <td>●</td> </tr> <tr> <td>P: Woman standing on a ladder fixing letters on a theater sign, while one woman watches and one man stands nearby.</td> <td></td> </tr> <tr> <td>H: Woman sleeps on a couch.</td> <td>●</td> </tr> <tr> <td>P: The Raiders complete the pass while the Dolphin defense scrambles to catch him.</td> <td></td> </tr> <tr> <td>H: The player is alone asleep in the bathtub.</td> <td>●</td> </tr> <tr> <td>P: Nine women in white robes with hoods walk on plush, green grass.</td> <td></td> </tr> <tr> <td>H: The woman is alone and asleep in her bedroom.</td> <td>●</td> </tr> <tr> <td>P: A little kid is jumping off a brick balcony.</td> <td></td> </tr> <tr> <td>H: A child is sleeping in bed.</td> <td>●</td> </tr> </table>	Premise / Hypothesis	Ours Applies	P: A woman in a white hat and red dress holding hands with a man in a black jacket with a flower in his upper pocket.		H: The woman is alone and asleep in the car on the moon.	●	P: Woman standing on a ladder fixing letters on a theater sign, while one woman watches and one man stands nearby.		H: Woman sleeps on a couch.	●	P: The Raiders complete the pass while the Dolphin defense scrambles to catch him.		H: The player is alone asleep in the bathtub.	●	P: Nine women in white robes with hoods walk on plush, green grass.		H: The woman is alone and asleep in her bedroom.	●	P: A little kid is jumping off a brick balcony.		H: A child is sleeping in bed.	●	Unit 9 – Our Non-Incremental Search IoU: 0.342 Explanation $(((\text{hyp:tok:sleeping} \wedge \text{neighbor:pre:tok:kid}) \vee (\text{hyp:tok:asleep} \wedge \text{neighbor:hyp:tok:in})) \vee \text{hyp:tok:sleeps})$ Samples selected by this explanation <table> <tr> <th>Premise / Hypothesis</th> <th>Inc. Applies</th> </tr> <tr> <td>P: A woman in a white hat and red dress holding hands with a man in a black jacket with a flower in his upper pocket.</td> <td></td> </tr> <tr> <td>H: The woman is alone and asleep in the car on the moon.</td> <td>●</td> </tr> <tr> <td>P: Woman standing on a ladder fixing letters on a theater sign, while one woman watches and one man stands nearby.</td> <td></td> </tr> <tr> <td>H: Woman sleeps on a couch.</td> <td>●</td> </tr> <tr> <td>P: The Raiders complete the pass while the Dolphin defense scrambles to catch him.</td> <td></td> </tr> <tr> <td>H: The player is alone asleep in the bathtub.</td> <td>●</td> </tr> <tr> <td>P: Nine women in white robes with hoods walk on plush, green grass.</td> <td></td> </tr> <tr> <td>H: The woman is alone and asleep in her bedroom.</td> <td>●</td> </tr> <tr> <td>P: A little kid is jumping off a brick balcony.</td> <td></td> </tr> <tr> <td>H: A child is sleeping in bed.</td> <td>●</td> </tr> </table>	Premise / Hypothesis	Inc. Applies	P: A woman in a white hat and red dress holding hands with a man in a black jacket with a flower in his upper pocket.		H: The woman is alone and asleep in the car on the moon.	●	P: Woman standing on a ladder fixing letters on a theater sign, while one woman watches and one man stands nearby.		H: Woman sleeps on a couch.	●	P: The Raiders complete the pass while the Dolphin defense scrambles to catch him.		H: The player is alone asleep in the bathtub.	●	P: Nine women in white robes with hoods walk on plush, green grass.		H: The woman is alone and asleep in her bedroom.	●	P: A little kid is jumping off a brick balcony.		H: A child is sleeping in bed.	●
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Figure 29: Comparison between incremental and non-incremental compositional explanations for Unit 9 in the LLaMA model.

Algorithm 1 Non-Incremental Beam Expansion

```

1: Input: Beam  $B_k$ , beam history  $B_{<k}$ , concept set  $\mathcal{L}^1$ , operator set  $\mathcal{O}$ , target arity  $k + 1$ 
2: Return: Candidate set  $\mathcal{B}_{k+1}$ 
3:  $\mathcal{B}_{k+1} \leftarrow \emptyset$ 
4: for each  $L \in B_k$  do
5:   for each  $c \in \mathcal{L}^1$  such that  $c \notin C(L)$  do
6:     for each  $\oplus \in \mathcal{O}$  do
7:        $L_{\text{inc}} \leftarrow L \oplus c$ 
8:        $\mathcal{B}_{k+1} \leftarrow \mathcal{B}_{k+1} \cup \{L_{\text{inc}}\}$ 
9:     end for
10:    for each  $\oplus \in \mathcal{O}$  do
11:      Identify the set  $\mathcal{S}_{\oplus}(L)$  of subformulas of  $L$  compatible with  $\oplus$ 
12:      for each  $S \in \mathcal{S}_{\oplus}(L)$  do
13:         $L_{\text{ni}} \leftarrow L[S \leftarrow S \oplus c]$ 
14:        if  $|C(L_{\text{ni}})| = k + 1$  then
15:           $\mathcal{B}_{k+1} \leftarrow \mathcal{B}_{k+1} \cup \{L_{\text{ni}}\}$ 
16:        end if
17:      end for
18:    end for
19:  end for
20: end for
21: for each  $L_1 \in B_k$  do
22:   for each  $L_2 \in B_{<k}$  do
23:     if  $|C(L_1) \cup C(L_2)| = k + 1$  then
24:        $\mathcal{B}_{k+1} \leftarrow \mathcal{B}_{k+1} \cup \mathcal{R}_{k+1}(L_1, L_2)$ 
25:     end if
26:   end for
27: end for
28: for each  $(L_1, L_2)$  in  $B_k$  do
29:   if  $|C(L_1) \cup C(L_2)| = k + 1$  then
30:      $\mathcal{B}_{k+1} \leftarrow \mathcal{B}_{k+1} \cup \mathcal{R}_{k+1}(L_1, L_2)$ 
31:   end if
32: end for
33: Return:  $\mathcal{B}_{k+1}$ 

```

Algorithm 2 Functional-Equivalence-Aware Beam Selection at Level k

```

1: Input: Ordered candidate list  $BC_k$ , activation mask  $N$ , concept masks  $\mathbf{M}$ , beam width  $b$ 
2: Return: Beam  $B_{k+1}$ , recombination candidates  $\mathcal{B}_{k+1}^{FE}$ 
3:  $B_{k+1} \leftarrow \emptyset$ 
4:  $\mathcal{B}_{k+1}^{FE} \leftarrow \emptyset$ 
5: for each  $L_j \in BC_k$  do
6:   if  $|B_{k+1}| = b$  then
7:      $L_{\min} \leftarrow \arg \min_{L \in B_{k+1}} IoU(N, \circ_L(\mathbf{M}))$ 
8:     if  $h(L_j) < IoU(N, \circ_{L_{\min}}(\mathbf{M}))$  then
9:       break
10:    end if
11:  end if
12:  Compute  $IoU(N, \circ_{L_j}(\mathbf{M}))$ 
13:  if  $|B_{k+1}| = b$  then
14:    if  $IoU(N, \circ_{L_j}(\mathbf{M})) < IoU(N, \circ_{L_{\min}}(\mathbf{M}))$  then
15:      continue
16:    end if
17:  end if
18:  Find  $L_t \in B_{k+1}$  such that  $\circ_{L_j}(\mathbf{M}) = \circ_{L_t}(\mathbf{M})$ , if it exists.
19:  if no such  $L_t$  exists then
20:    if  $|B_{k+1}| < b$  then
21:       $B_{k+1} \leftarrow B_{k+1} \cup \{L_j\}$ 
22:    else
23:       $B_{k+1} \leftarrow (B_{k+1} \setminus \{L_{\min}\}) \cup \{L_j\}$ 
24:    end if
25:  else if  $L_j$  and  $L_t$  have Type-1 or Type-2 equivalence then
26:    if  $L_t$  is unverified and  $L_j$  is verified then
27:       $B_{k+1} \leftarrow (B_{k+1} \setminus \{L_t\}) \cup \{L_j\}$ 
28:    end if
29:  else if  $L_j$  and  $L_t$  have Type-3 equivalence then
30:    if  $|B_{k+1}| < b$  then
31:       $B_{k+1} \leftarrow B_{k+1} \cup \{L_j\}$ 
32:    else
33:       $B_{k+1} \leftarrow (B_{k+1} \setminus \{L_{\min}\}) \cup \{L_j\}$ 
34:    end if
35:     $\mathcal{B}_{k+1}^{FE} \leftarrow \mathcal{B}_{k+1}^{FE} \cup \mathcal{R}_t(L_j, L_t)$ 
36:  end if
37: end for
38: Return:  $B_{k+1}, \mathcal{B}_{k+1}^{FE}$ 

```

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